DIVINE EDUCATION CENTRE

INTEGRATED SCIENCE

LESSON NOTES TERM ONE 2026

0784540287/0751565742

PRIMARY FOUR

THEME :THE WORLD OF LIVING THINGS

TOPIC 1 : PLANT LIFE

- ✓ Flowering and non flowering plants
- ✓ Plant roots
- ✓ Types of roots
- ✓ Plant stems
- ✓ Plant leaves
- ✓ Photosynthesis
- ✓ Transpiration
- ✓ Flowers
- ✓ Pollination in flowering plants
- ✓ Seeds
- ✓ Seed germination
- ✓ Uses of plants to people
- ✓ End of topic Revision exercise

THEME :THE WORLD OF LIVING THINGS

TOPIC 2 : GROWING CROPS

- ✓ Common crops
- ✓ Classification of crops
- ✓ Garden tools,equipment and materials
- ✓ Crop growing practices
- ✓ Caring for crops
- ✓ Mulching
- ✓ Preserving crops
- ✓ Crop pests and diseases
- ✓ End of topic revision exercise

THEME : OUR ENVIRONMENT

TOPIC 3 : WEATHER CHANGES AROUND US

- ✓ Changes in weather
- ✓ Rainfall
- ✓ Water cycle
- ✓ Wind
- ✓ Weather instruments
- ✓ Meaning of other key terms in weather changes around us
- ✓ End of topic revision exercise

THEME :HUMAN HEALTH

TOPIC 4 : PERSONAL HYGIENE

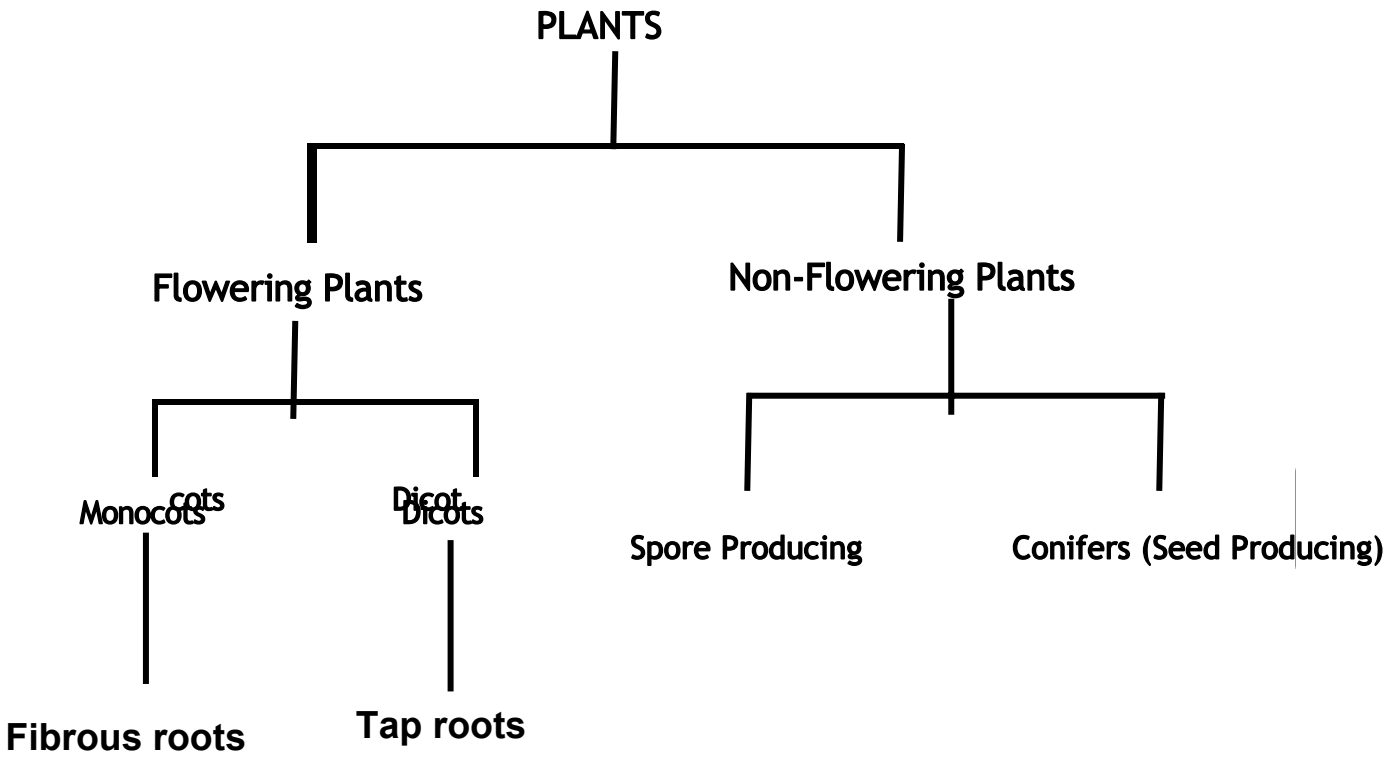
- ✓ Parts of the body to be kept clean
- ✓ Things used to keep our body clean
- ✓ Dangers of failing to maitain personal hygiene
- ✓ Signs of poor personal hygiene
- ✓ Keeping bedding and clothes clean
- ✓ Roles of School Health clubs in promoting personal hygiene
- ✓ End of topic revision exercise

Week one

Lesson 1

CLASSIFICATION OF PLANTS

Classification means grouping according to similar characteristics



Non flowering plants

1. What are non-flowering plants?

. Non flowering plants are plants that do not bear flowers

conifers

. fir
. cedar
. pines
. cypress
. podo

spore producing

. ferns

. mosses

. liverwort

horsetails

ACTIVITY

1. **How do flowering plants reproduce?**
2. **State two groups of flowering plants.**
3. **Write down two examples of spore producing plants.**

Lesson two

FLOWERING PLANTS

Flowering plants are plants that bear flowers

EXAMPLES OF FLOWERING PLANT

Beanplant

. Peas plant

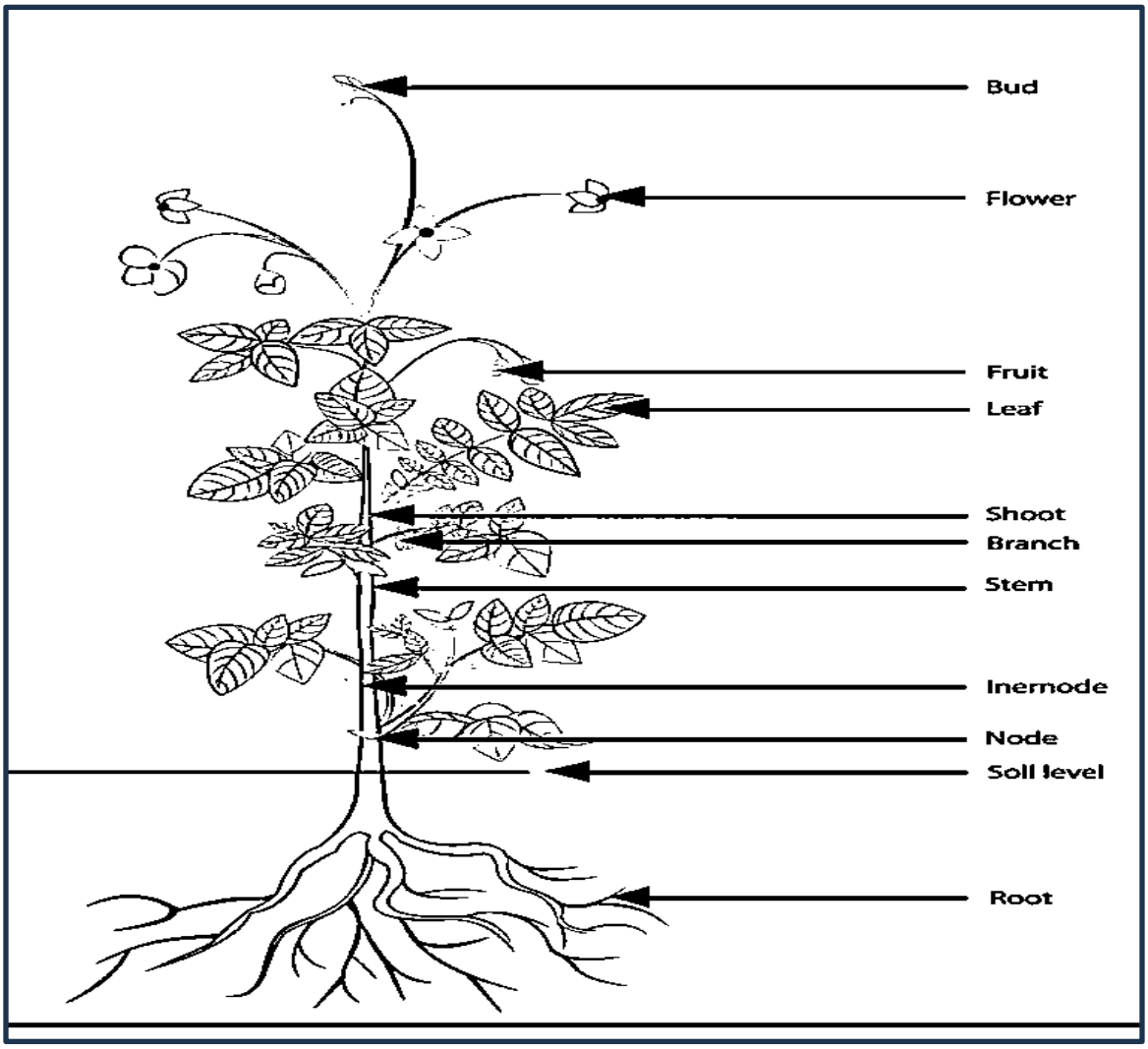
. Maize plant

. Groundnuts plant

. Millet plant

. rice plant etc

Parts of a flowering plant.



ACTIVITY

1. Give two examples of flowering plants
2. Why is a rice plant regarded as a flowering plant?
3. How do non flowering plants reproduce?

Lesson three

SYSTEMS OF A FLOWERING PLANT

2. The flowering plant has two systems namely:

a) Shoot system

b) Root system

Root system.

Part of the plant that grows downward into the soil.

Parts of the root system

. Taproot

. Root cap

. Root hair

. Lateral roots

ROOTS

1 .This is the part of a plant which grows in the soil

2. A true root systemdevelops fromthe radicle of the embryo.

Importance of roots to plants

. Roots hold the plant firmly into the soil.

. Roots absorb water and mineral salts fromthe soil .

. Some plants store food for the roots. E.g. cassava roots, sweet potato roots, carrot roots etc.

Importance of roots to man

a) Some roots provide man with food. e.g. Cassava, Sweet potatoes and Carrots.

b) Some roots are used as herbal medicine.

c) Some roots are used to make craft items.

d) Roots are used as wood fuel

e) Some roots are sold to get money

Note:

Roots that store food are called root tubers.

Capilary attraction is the process by which soluble minerals/plant nutrients are absorbed from the soil through roots to the entire plant.

Types of roots

- primary roots
- secondary roots

ACTIVITY

1. **Writ e two systems of a flowering plant**
2. **Name one part of a root system**
3. **How are roots useful to man?**

Lesson 4

PRIMARY ROOTS

What are primary roots?

Primary roots are roots that develop fromthe radicle.

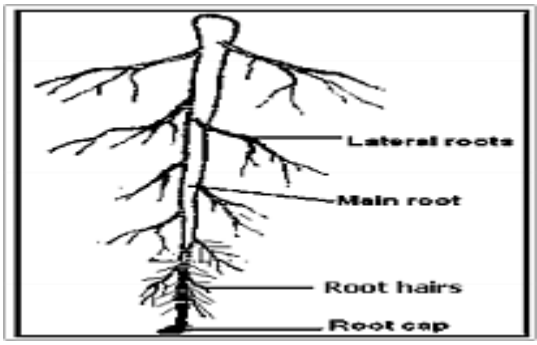
EXAMPLES OF PRIMARY ROOTS

- Tap root
- Fibrous roots

TAP ROOT

These are roots fromwhich lateral roots develop

Structure of a taproot



Examples of plants with taproots

a) Bean plants

b) Soya

c) Groundnuts

d) Mango trees

e) Tomatoes

f) Guava tree

Functions of parts of roots

a. Roothair

To absorb water and mineral salts

b. Root cap

To protect the tip of the root.

c) Lateral root

Holds the plant into the soil

NOTE

Primary roots are divided into two root systems

. Taproot system

. Fibrous root system

Activity

Draw the root systems above

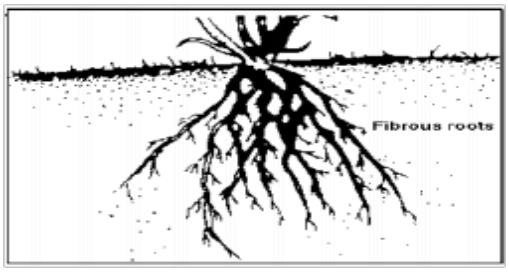
Fibrous roots

- This is a type of root systemwhere there are many roots growing randomly.

. .There is no main root.

- They don'tgo deep into the ground.

Structure of fibrous roots



Examples of plants with fibrous roots

. Maize

. Sorghum

. Sugar cane

. Millet

. Rice.

Beanplant

. Peas plant

. Maize plant

. Groundnuts plant

. Millet plant

. rice plant etc

ACTIVITY

1. **What are primary roots?**
2. **Draw a structure of tap root system**
3. **Write two examples of plants with fibrous roots**

Lesson 5

Secondary roots

These are roots that develop fromthe stems or leaves of a plant

Secondary roots are subdivided into two:

- Adventitious rootsb
- Aerial roots

AERIAL ROOTS

These are roots that appear above the ground eg

- Prop roots
- Clasping roots
- Buttress roots
- Stilt roots
- Breathing roots

PROP ROOTS

1 . They develop fromnodes near the ground level.

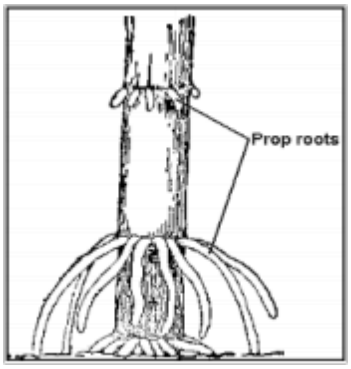
What is the main function of prop roots?

These are roots which give extra support to plants.

Examples of plants with prop roots:

- Maize
- Sorghum
- Sugar cane
- millet
- rice
- wheat
- elephant grass etc.

Structure of prop roots



Storage roots/ Root tubers

These are swollen underground ***rots***with stored food.

Examples of storage roots:

- Cassava
- Sweet potatoes
- Carrots
- beet root
- root turnip
- Swedes

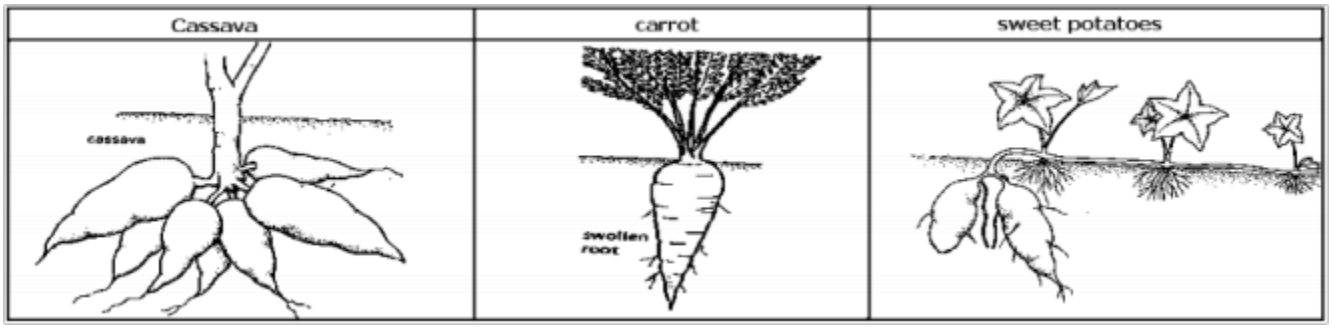
sugar beet

. white yams

. parsnips

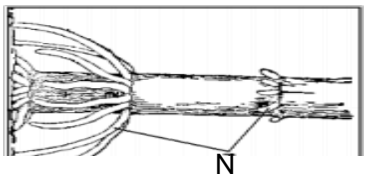
. dahlia

Structures of storage roots



ACTIVITY

1. **How are secondary roots useful to a plant?**
2. **Below is a structure of a root system. Use it to answer questions below**



Identify the root system marked N.

Lesson 6

Shoot system

1 .Part of the plant that grow above the ground.

Parts found on the shoot system.

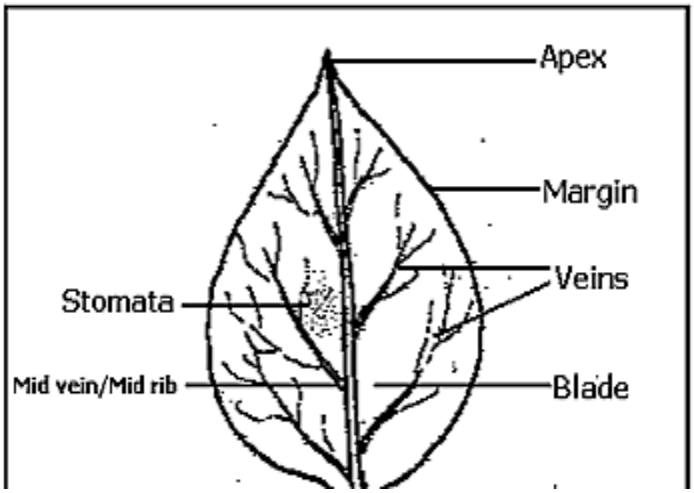
. Stem,
. Leaves,
. Lateral buds,
. Terminal bud,

Fruit,

. Flower,
. Node.
. Internodes

THE LEAVES OF A PLANT

Parts of a leaf



Functions of some parts of a leaf

Mid vein

. Transports water and mineral salts fromthe leaf stalk to the leaf.

. Transports starch fromthe leaf to the leaf stalk.

Veins

. Distributes water and mineral salts in the leaf

Leaf stalk

Transports starch to the lower parts of the plant

. Transports water to the leaf.

. It attaches the leaf to the stem

Stomata

. It is used for breathing.

. It helps in transpiration

LAMINA

Makes food

Uses of leaves to a plant

. Leaves make starch

Plants store food in leaves

. Leaves help the plant to breathe

. Leaves carry out transpiration.

. Some leaves are used for propagation e.g bryophyllum leaf.(ekiyondo)

Note:

The main function of a leaf to a plant is to make food.

Uses of leaves to man

. Some leaves are used as food.

. Some leaves are used as herbal medicine to cure disease.

. Some leaves are used to thatch houses.

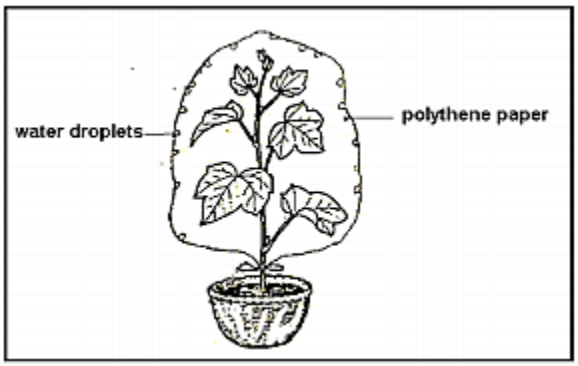
. They are used to mulch gardens.

PROCESSES THAT TAKE PLACE IN LEAVES

. Transpiration

Transpiration is the process by which plants lose water in form of v apour to the atmosphere.

An experiment to show that plants lose water



REQUIREMENTS

. Transparent polythene paper

. Plant under sunshine

. Rubber band. Leave it there for two hours. Onions. Some leaves can be used as decorations

. Some leaves can be used in making craft materials.

. Some leaves can be used as costumes i.e. cultural dances like Imbalu dance.

Plants that

. Photosynthesis

TRANSPIRATION store food in their leaves

. Cabbages

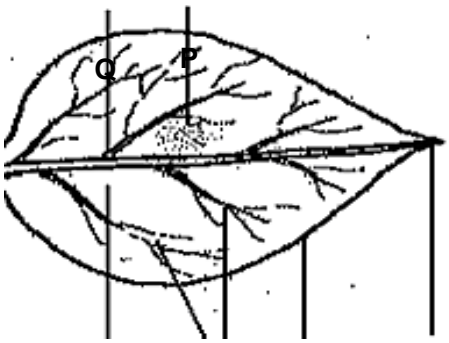
. Sisal plants

. Garlic

. Aloevera

Activity

1. **The diagram below is of a leaf. Use it to answer questions that follow.**



Name the parts marked

P----------

Q----------

R----------

2. **Write down two partsof a shoot system**
3. **Suggest two uses of leaves to a plant**
4. **What is transpiration?**
5. **Write two factores that affect the rate of transpiration**
6. **How does humidity affect the rate of transpiration?**

Week two

Lesson 1

STEPS

. Wrap apolythene paper around a shoot

. Tie a rubber band to prevent the escape of vapour

. Observe the inside the polythene paper

Results

Drops of water are found on the inside of the polythene paper.

Importance of transpiration to plants

. Transpiration helps to cool the plants

. Transpiration gives room for the plant to absorb more water.

IMPORTANCE OF TRANSPIRATION TO THE ENVIROMENT

It helps in rainformation

DANGERS OF TRANSPIRATION

. Excessive transpiration results into wilting.

. It lowers crop yields

Note: Wilting is the process through which the rate of water absorption is less than water loss in plants.

FACTORS THAT AFFECT THE RATE OF TRANSPIRATION

. **Temperature:**

When temperature is high transpiration is high and when it is low it lowers

. **Humidity**

When humidity is high transpiration I low and when it is low transpiration is high

. **Size of the leaf**

A small leaf loses less water than a big leaf

Wind

When the speed of wind is high transpiration is high

. **sunlight intensity**

Bright light fromthe sun increases transpiration

HOW PLANTS CONTROL THE RATE OF TRANSPIRATION

. By shedding leaves

. By developing thorns

. By developing needle like leaves

. By folding their leaves

. By developing a layer of wax

Activity

1. **Write two importances of transpiration to :**

   (i) people

   (ii) plants

2. **How is transpiration dangerous to the environment?**

Lesson 2

PHOTOSYNTHESIS

. Photosynthesis is the process by which plants make starch.

- Photo means light
- Synthesis means to make

. The food made during photosynthesis is starch.

Conditions necessary for photosynthesis:

- Chlorophyll
- Sunlight.

Chlorophyll:

- This is the green colouring matter found in leaves.
- Its main function is to trap sunlight energy fromthe sun.

Sunlight

It splits water into hydrogen and oxygen

Raw materials of photosynthesis

Water

Combines with carbon dioxide to formstarch.

Carbon dioxide:

This gas passes through the small holes on a leaf called stomata.

Carbon dioxide combines with water to formstarch.

Requirements for photosynthesis

- Chlorophyll
- Sunlight
- Water
- Carbon dioxide

The products of photosynthesis.

- Starch - end product
- Oxygen - by product given off

Importance of photosynthesis to plants.

Plants get food fromphotosynthesis.

Importance of photosynthesis to animals.

. Animals get food fromphotosynthesis.

. Animals get oxygen fromphotosynthesis.

Importance of photosynthesis to the environment.

It reduces carbon dioxide in the atmosphere.

. Simple lobbed leaf.

Activity

1. **Give the meaning ogf the term photosynthesis.**
2. **Write two conditions necessary for photosynthesis.**
3. **State the importance of water as a raw material needde for photosynthesis to take place.**
4. **How do people benefit from the process of photosynthesis.**

Lesson 3

TYPES OF LEAVES.

. Simple leaves

. Compound leaves.

Simple leaves

A simple leaf is a leaf with one leaf let on one leaf stalk

Examples of simple leaves:

. Simple serrated leaf

simple digitate leaf.

. simple divided entire leaf.

Structures of simple leaves

Simple entire leaf.



Simple serrated leaf .



simple digitate



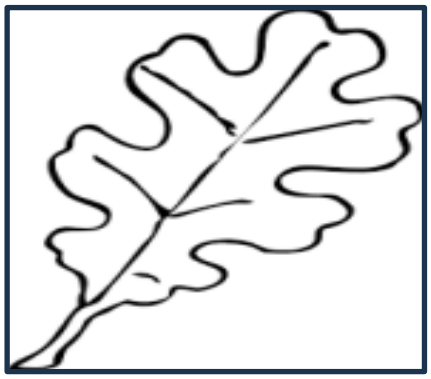
simple palmete leaf



simple divided leaf



simple lobed leaf



Activity

1. **What are simple leaves?**
2. **Mention two types of leaves**
3. **in the space provided below, draw a structure of a simple ventilated leaf.**



Lesson 4

Compound leaves

A compound leaf is a leaf with many leaflets on one leaf stalk

Note:

The leaf blade is completely divided to formleaflets.

Examples of compound leaves:

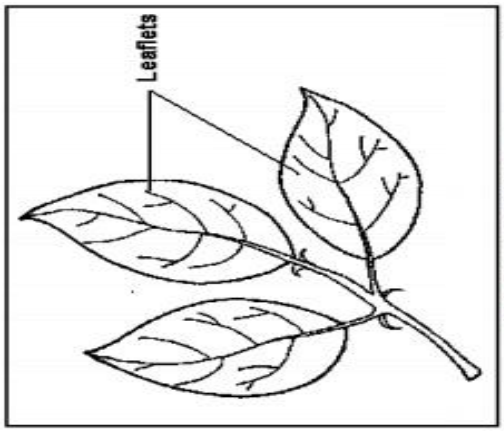
. Compound trifoliate

. Compound pinnate

. Compound bipinnate

Structures of compound leaves

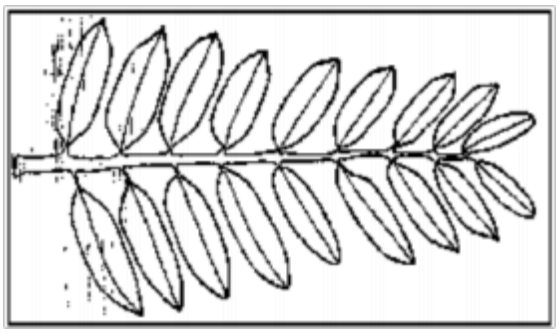
Compound trifoliate.



Compound bipinnate



Compound pinnate.



Activity

1. **Define the term compound leaves**
2. **Give two examples of compound leaves**
3. **In the space provided below, draw a diagram of compound trifoliate leaf**

Lesson 5

LEAF VENATION

Leaf venation is the arrangement of veins in a leaf.

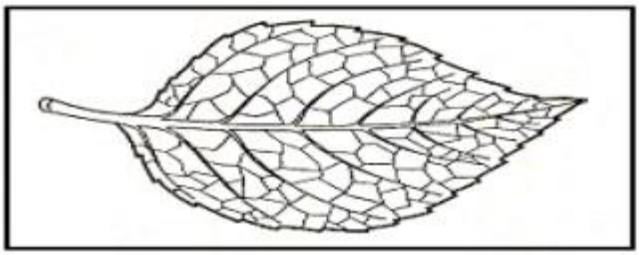
Types of leaf venation

. Network Venation

. Parallel Venation

The network venation

In network venation the veins make a net like structure

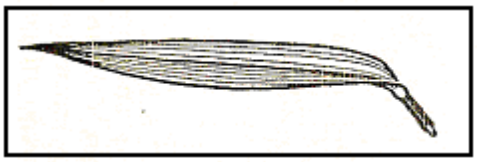


Examples of plants with network veins.

- Beans
- Peas
- Hibiscus plant.
- Groundnuts
- Mango plant and many tree plants.

The parallel venation

In parallel venation veins run fromthe stalkto the apex



Examples of plants with parallel veins

. Maize
. Sugar cane
. Millet
. Wheat
. Rice
. Elephant grass etc.

Activity

1. **Give the meaning of the term leaf venation**
2. **Give two types of leave venation**
3. **Draw a structure of parallel leaf venation**



4. **Mention any two examples of plants with network veins**

Lesson 6

STEMS:

These are parts of a flowering plant with buds.

Functions of a stemto plants

. It holds plants leaves and branches.
. It transports water and mineral salts fromroots to the leaves.
. It transports food fromleaves to other parts of the plant.
. Some plants store food in stems e.g. sugarcane, ginger, yams, irish potato etc.

. Green stems make food for the plant.

1 . Uses of stems to man

. Many stems are used as fuel.

. Some are used as herbal medicine.

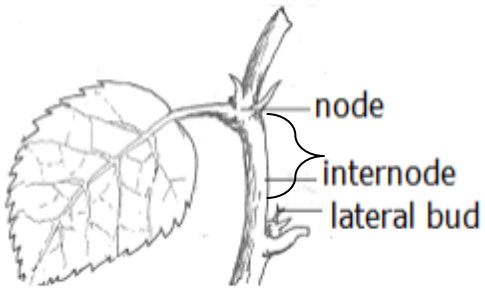
. Some stems are used as food. e.g. Sugar canes, Ginger plant, Irish potatoes, etc

. Some stems area source of timber.

. Stems are used as poles.

. Some stems are sold to get money

Parts of a stem



Terminal bud is the growing tip of a plant.

- **The lateral bud** can grow into a branch or flower.
- A **node** is the part of a stemwhere a leaf is fixed.
- An **internode** is the distance (region) between two successive nodes.

Note: An axil isa space between each leaf and the stem.

Activity

1. **What are stems?**
2. **How are stems useful to a plant?**
3. **Give two uses of stems to man**

WEEK THREE

Lesson 1

TYPES OF STEMS

- Upright stem/ erect stem
- Underground stems
- Weak stems.

Upright/erect stems

They are strong stems that are able to stand upright without any support **Examples of plants with Upright/erect stems**:

- Treese.g Muvule etc.
- Jackfruits
- Mangoes
- Maize etc.

Weak stems.

These are stems which cannot support themselves upright.

Identify the two groups of weak stems

- Climbing stems
- Creeping stems

ACTIVITY

1. **Write two types of stems**
2. **Name two examples of plants with upright stems.**
3. **Identify two groupd of week stems.**

Lesson 2

CLIMBING STEMS

They are weak stems that enable themto climb

Why do plants climb others.

- Plants climb others to get enough sunlight.
- For support

How weak stems climb others.

. By use of tendrils e.g. passionfruits, pea plant and some beans etc.

. By twining e.g. some yams, some beans. Morning glory money plant,guard, oilnuts, loofah pumpkins cucumber vanilla.

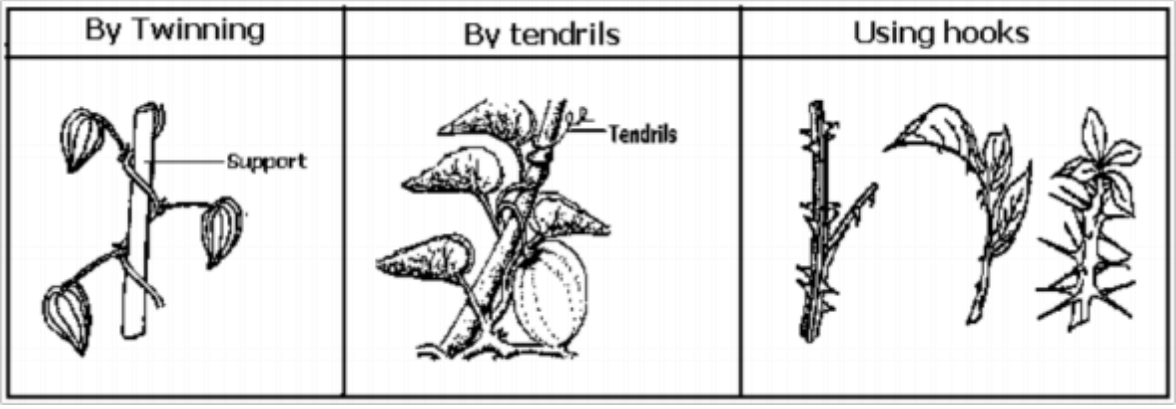
. By use of hooks & thorns e.g.bougain villea

Structures that help plants with weak stems to climb

. Tendrils

. Hooks or thorns

Illustration of climbing stems



Activity

1. **Why do plants climb others?**
2. **Write two structure plants with weak stems use to climb others.]\\=**

Lesson 3

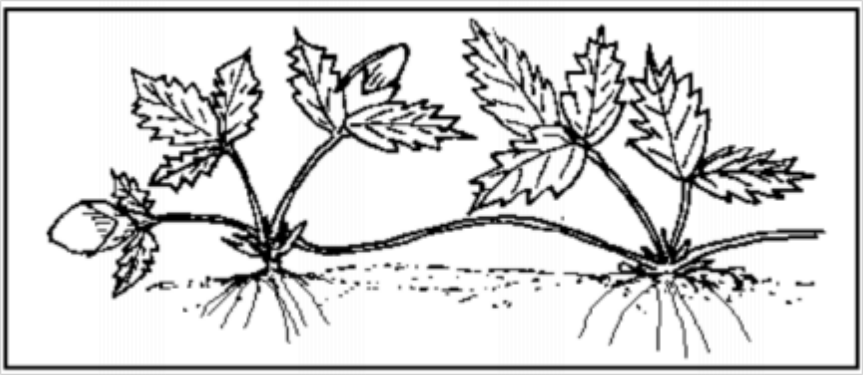
CREEPING STEMS.

These are stems that run along the ground.

Examples of plants with creeping stems.

- Water melon
- Pumpkins
- Sweet potatoes
- Cucumber
- Straw berry

Illustration of creeping stems.



UNDERGROUND STEMS OR STORAGE STEMS.

These are stems that mainly grow underground.

Types of underground stems

- Stemtubers
- Bulbs
- Rhizomes
- Corms

Lesson 4

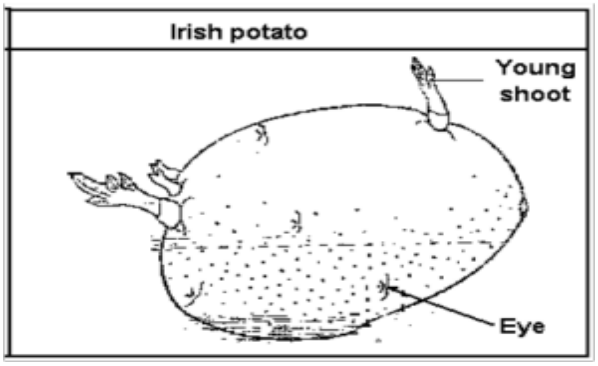
STEMTUBERS

- These are swollen underground stems with stored food.
- We eat themas food.

Examples of stemtubers:

a) Irish potatoes.

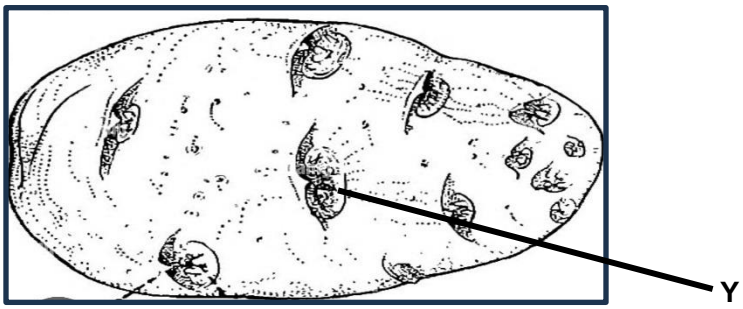
Structures of stemtubers



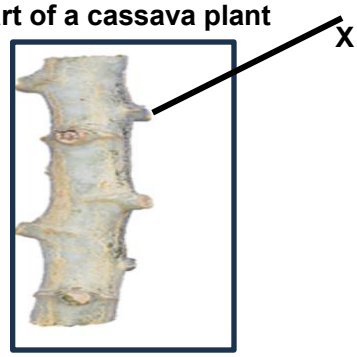
Activity

The diagram below shows part of a cassava plant and irish potato that are used for propagation.Study and use them to answer questions that follow.

irish potato



part of a cassava plant



1. **Name the part of a cassava plant marked X.**

2. State the method of vegetative propagation where the part of a cassava plant shown above is used.
3. In which one way is the function of part x of a cassava plant similar to that of a y of the irish potato.
4. Give the importance of the tuber in the propagation of an irish potato.

Lesson 5

BULBS

- A bulb is an underground stem.
- Food in a bulb is stored in the fleshy leaves.
- Bulbs have adventitious roots.

Examples of bulbs:

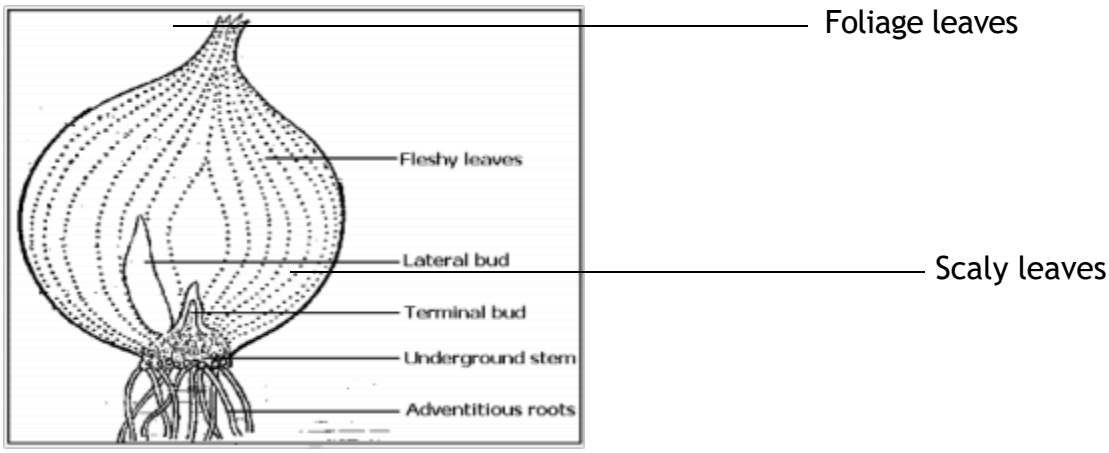
. Onions

. Garlic

. Leek

. Shallots

A bulb of an onion



. **Buds** –Develop into a shoot system

. **scaly leaves** –protect the fleshy leaves

. **foliage leaves** – Transports water and mineral salts from the roots to the leaves.

. **stem** - holds the leaves

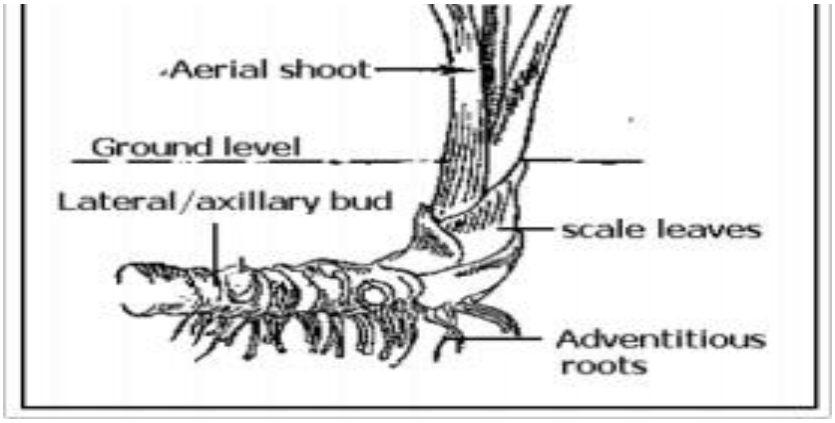
. **storage leaves / fleshy leaves**- sore the made food.

. **Adventitious roots** –Absorb water and mineral salts

RHIZOMES

- A rhizome is a horizontal underground stem
- Rhizomes develop adventitious roots
- Rhizomes store food in the stem
- An example of a rhizom is ginger, cannalily, turmeric

Structure of a rhizome



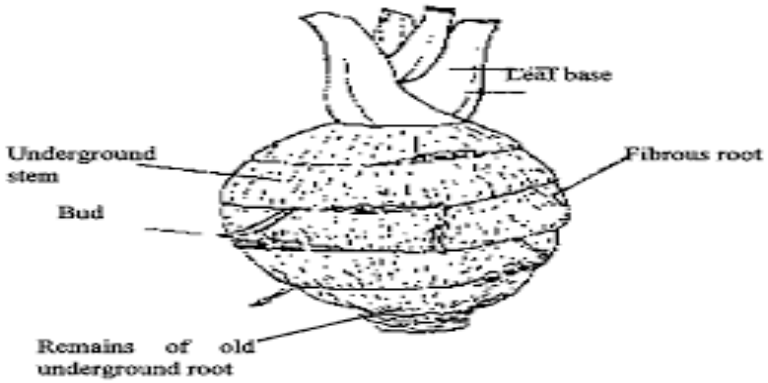
CORMS:

- o A corm is a short vertical underground stem.
- o It is swollen with stored food
- o It has scale leaves, lateral buds and adventitious roots.

Examples of corms:

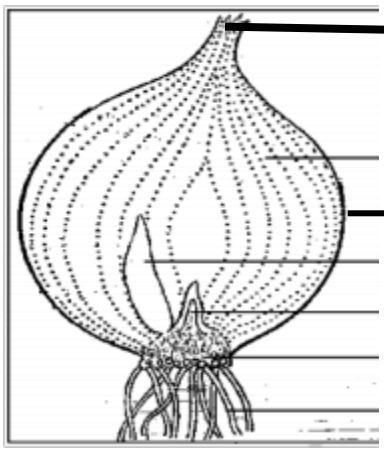
- . Coco-yams
- . Crocus
- . gladiolus

STRUCTURE FOR COCO YAMS



Activity

The diagram below shows an onion plant. Study and use it to answer questions that follow.



A

B

C

1. **Name the parts of an onion marked B and C.**
2. **Give the importance of the part marked A to an onion**
3. **State the importance of the part marked D to an onion**

Week 4

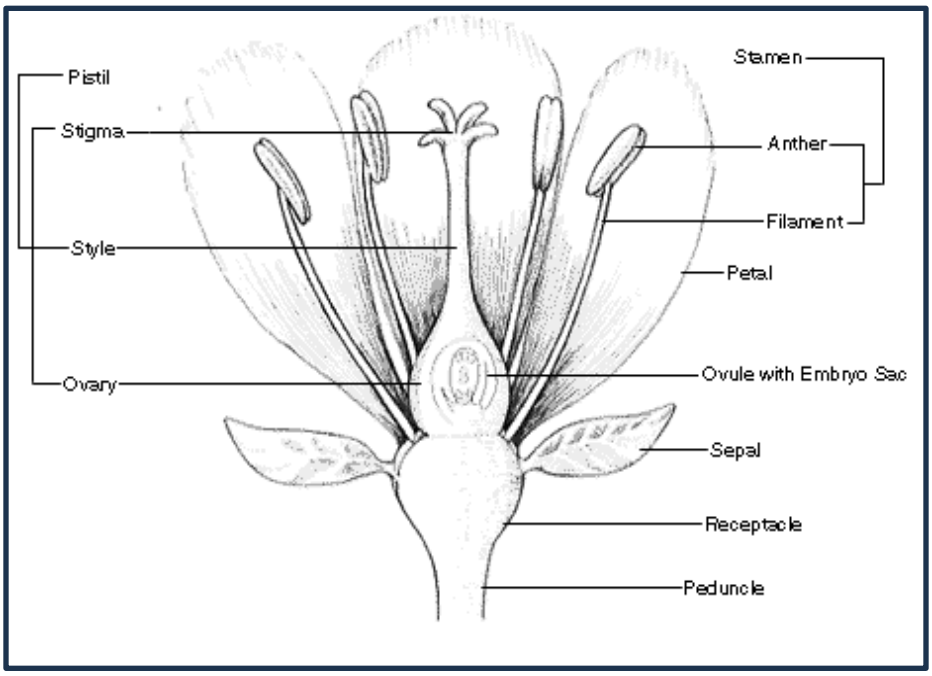
Lesson 1

THE FLOWER

. A flower is a reproductive part of a plant.

. The main function of a flower is for reproduction

Parts of a flower



FUNCTIONS OF THE PARTS OF A FLOWER.

Sepals

. They protect the flower when still in a bud stage.

. They make food for the plant.

Note

A group of sepals is called **calyx**

Petals

. Petals attract pollinators

. They are coloured parts of a flower.

Note

A group of petals is called **corolla.**

Filament

. Holds the antherhead in position.

Anthers

. Produce and store pollen grains.

Stigma

. Receives pollen grains .

Style

. Holds the stigma in position.

. Joins the stigma to the ovary

. It passes pollen grains to the ovary.

Ovary

- Produces ovules
- Stores ovules
- It is where fertilization takes place in a flower

Flower stalk

- The stalk attaches the flower on the stemor branch

Receptacles

- It holds the ovary,nectaries,sepals,and petals.
- It produces nectar

THE PISTIL AND STAMEN

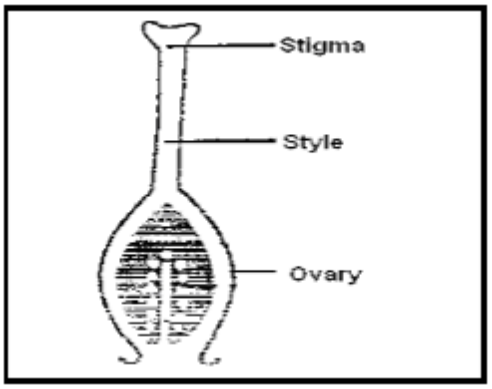
The Pistil/carpel

This is the female part of a flower.

Parts that make up the pistil:

- Stigma
- Style
- Ovary

Structure of the Pistil



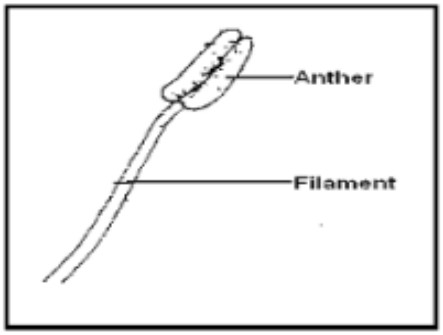
The Stamen

This is the male part of a flower.

Parts that make up the stamen:

- Anther
- Filament

The structure of the stamen



REPRODUCTIVE CELLS IN PLANTS

How do we call each of the following reproductive cells in plants?

- Female cells : Ovules
- Males cells : Pollen grains

Activity

1. **Name the part of the flower where fertilization takes place.**
2. **State the function of flowers to a plant.**
3. **Give any one reason why the petals of some flowers are brightly coloured.**

4. **Identify the part of a flower that develops into:**
   - **(i) fruit after fertilisation**
   - **(ii) seeds after fertilisation**
5. **State one way in which the stigma of a flower is adopted to its function.**
6. **State the difference between the stamen and pistil of a flower.**

Lesson 2

POLLINATION

Pollination is the transfer of pollen grains fromthe anthers to the stigma of a flower on the plant.

Agents of pollination:

These are things responsible for the transfer of pollen grains to the stigma.

Agents of pollination

- Wind
- Running water or flowing water
- Animals
- Birds
- insects (wasps, bees, butterflies, moth)

Types of pollination

- Self-pollination
- Cross pollination

Self pollination

. Self-pollination is the transfer of pollen grains from the anthers to the stigma of the same flower.

. Flowers that under self-pollination have filaments taller than the style.

EXAMPLES OF PLANTS THAT UNDERGO SELF POLLINATION

. Beans

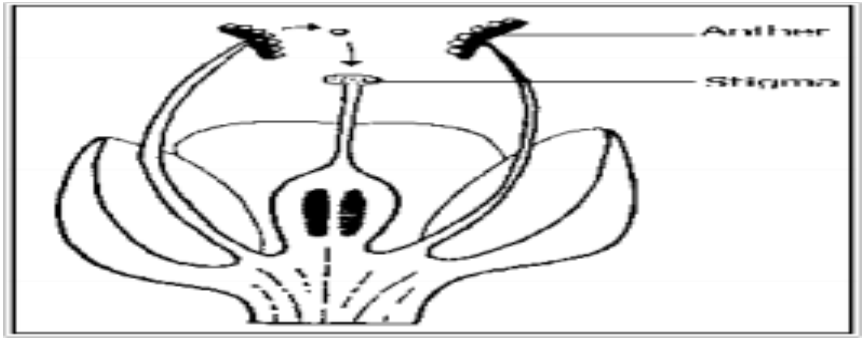
. Groundnuts

. Mangoes

. Oranges

. Cowpeas etc

DIAGRAMSHOWING SELF POLLINATION



Cross-pollination

. Cross-pollination is the transfer of pollen grains fromthe anther of one flower to the stigma of another flower on different plants but of the same kind.

. Flowers that undergo cross pollination have filaments shorter than the styles.

EXAMPLES OF PLANTS THAT UNDER CROSS POLLINATION

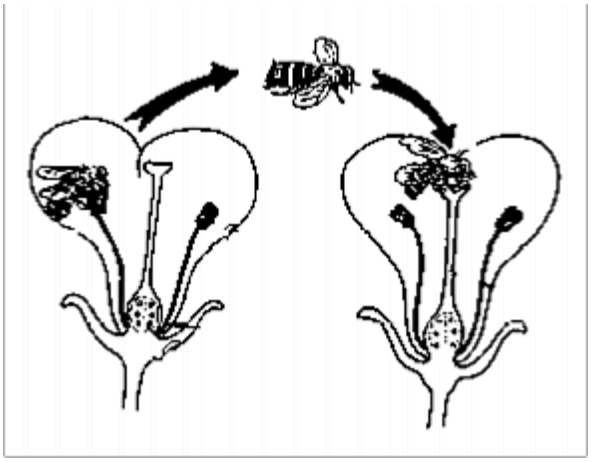
. Maize

. Passionfruit

. Coconut

. Pawpaw

DIADRAMSHOWING CROSS POLLINATION



Characteristics of insect pollinated flowers.

- Have brightly coloured petals
- Have nectar
- Have good smell/ scent Produce few pollen
- Produce sticky pollen grains

Characteristics of wind pollinated flowers.

- Produce a lot of pollen grains
- Have dull petals
- Have no nectar
- No smell / scent
- The pollen grains are light

Importance of flowers to man

- Flowers are used for decoration on various functions.
- Some flowers are used in making of perfumes.
- Some flowers are used in the making of insecticides e.g. Pyrethrum.
- They are used to make dyes.

. Some flowers area source of food.

. IT is given as gifts

. They are used to pay last respect to the dead

Importance of flowers to the plant

Flowers produce fruits and seeds that help in reproduction

Some flowers store food

Activity

1. **Apart from having bright colours, state any one other thing that attract pollinators to flowers.**
2. **Give the reason why a moth is able to pollinate flowers at night.**
3. **Name two types of pollination in flowering plants.**
4. **State two differences between insect pollinated flowers and wind pollinated flowers.**
5. **Apart from insects, name any one other agent of pollination.**
6. **State one characteristic that attract bees to visit flowers**

Lesson 3

FERTILIZATION

- Fertilization is the union of the female and male reproductive cells to forma zygote.
- Fertilization in plants takes place in the ovary.

In flowering plants, the male gametes are the pollen grains and the female gametes are the ovules.

- After fertilization, the style dies and falls off

Note: After fertilization, the ovules develop into seeds and the ovary develops into a fruit.

Fertilization takes place in the OVARY

A FRUIT

A fruit is a fertilized ovary

Examples of fruit

- **mangoes**
- **bananas**
- **oranges**
- **pineapples**
- **tomatoes**
- **pawpaws**

IMPORTANCE OF FRUIT TO MAN

. They are sold to get money

. They are eaten as food

Importance of fruit to a plant

- Protect seeds
- Help in scattering of seeds

SEEDS

- A seed is a fertilized mature Ovule.
- A seed develops into a new plant called seedling

Importance of seeds to man

- They are used for propagation.
- We eat seeds e.g. Beans, maize grain, groundnuts, etc.
- Some seeds are used to make decorations.
- They are used as herbal medicine.

Types/ Classes of seeds

- Dicotyledonous seeds (dicots)
- Monocotyledonous seeds (monocots)

Dicotyledonous seeds

- These are seeds with two cotyledons.
- Di means two.

Examples of Dicotyledonous seeds:

Bean seeds	Pea seeds
Ground nut seeds	Soya bean seeds

Coffee seeds

Avocado seeds

Pumpkin seeds

Oil nuts

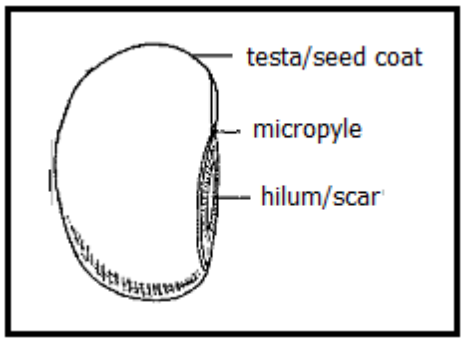
6. Mango seeds

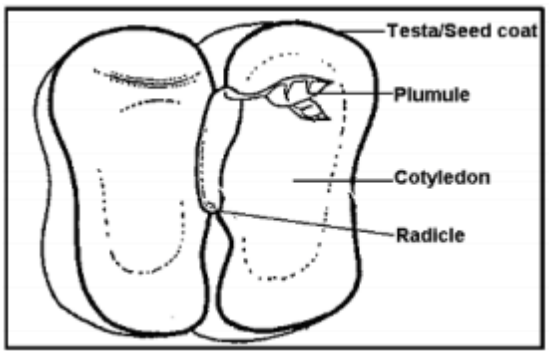
Orange seeds

Water melon seeds

Simsim

External and internal parts of a bean seed.





Functions of parts of a bean seed

Testa/ seed coat

. It protects the inner parts of a seed

Micro Pyle

. IT allows oxygen and air and water into a seed during germination.

Cotyledon

. It stores food for the embryo in dicots.

Note: During germination, the cotyledon provides food to the embryo.

Radicle

. It develops into a root systemduring germination

Plumule

. It grows into the shoot systemduring germination.

EMBRYO

. It grows into a seedling

. It is made up of two parts: plumule and radicle

Monocotyledonous seeds

- These are seeds with one cotyledon.
- Mono means one

Examples of monocotyledonous seeds:

. Maize

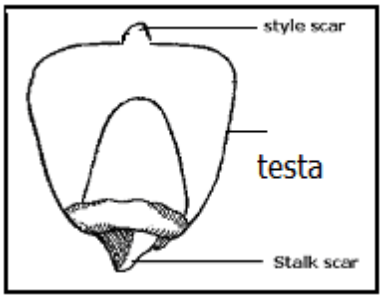
. Millet

. Sorghum

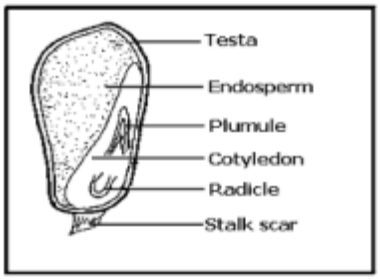
. Rice

. Whea
. Barley
. Oat
. Rye

External parts of a maize grain



The internal parts of a maize grain



Functions of parts of a maize grain

Testa/ seed coat

. It protects the inner parts of a seed

Cotyledon

. Absorbs and supplies food fromthe endospermto the embryo

Radicle

. It develops into a root systemduring germination

Plumule

It grows into the shoot systemduring germination.

Endosperm

. It stores food for the embryo in monocots

Stalkscar

. **It attaches the grain to a cob**

Difference between monocots and dicots.

Dicotyledonous	Monocotyledonous
1 . Have two cotyledons Most Dicotyledonous seeds undergo epigeal germination 3. Stores food in the cotyledon. 4. Have taproot system 5. Have networkveins	-Have one cotyledon Most-Monocots undergo hypogeal germination -Stores food in the endosperm. -Have fibrous root system -Have parallel veins

Lesson 4

GERMINATION

. Germination is the growing of a seed into a young plant.

. A young plant is called a seedling.

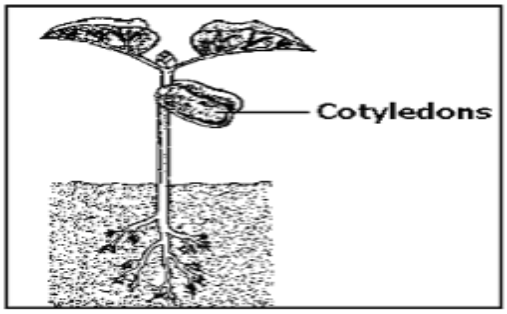
Types of germination

- o Epigeal germination
- o Hypogeal germination.

Epigeal germination:

. Epigeal germination is a type of germination where the cotyledons come out of the soil.

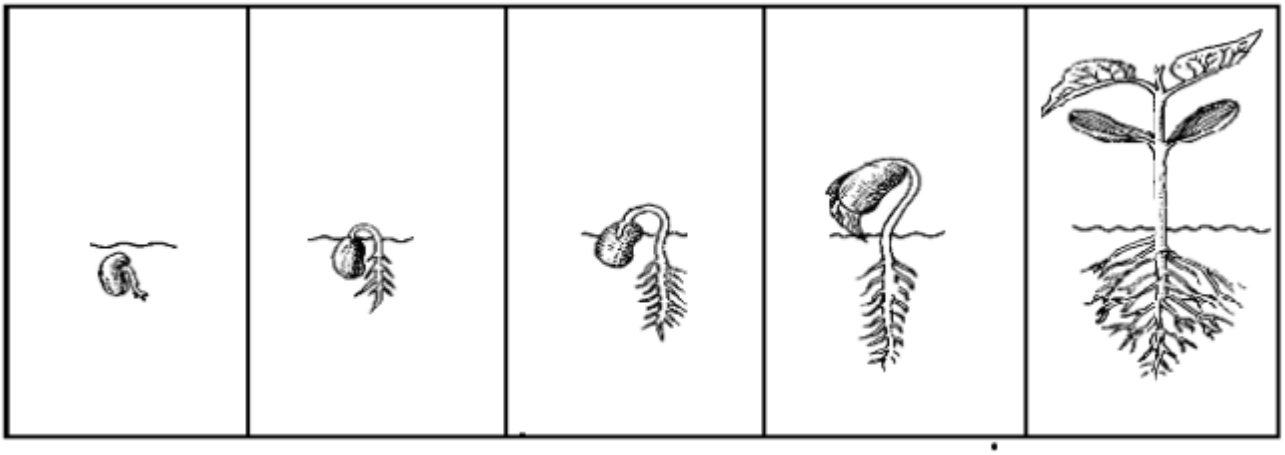
An illustration of epigeal germination.



Examples of seeds that carry out epigeal germination.

. Beans seeds
. Soya bean seeds
. Groundnut seeds
. Coffee seeds
. Orange seeds
. Mango plants
. Pumpkin seeds
. Watermelon seeds
. Cow peas

Stages in epigeal germination



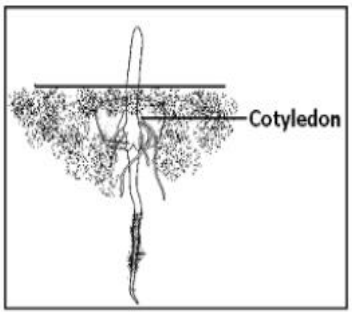
Hypogeal germination

1. Hypogeal germination is a type of germination where the cotyledon remains

under the ground.

2. All monocotyledonous seeds carry out hypogeal germination

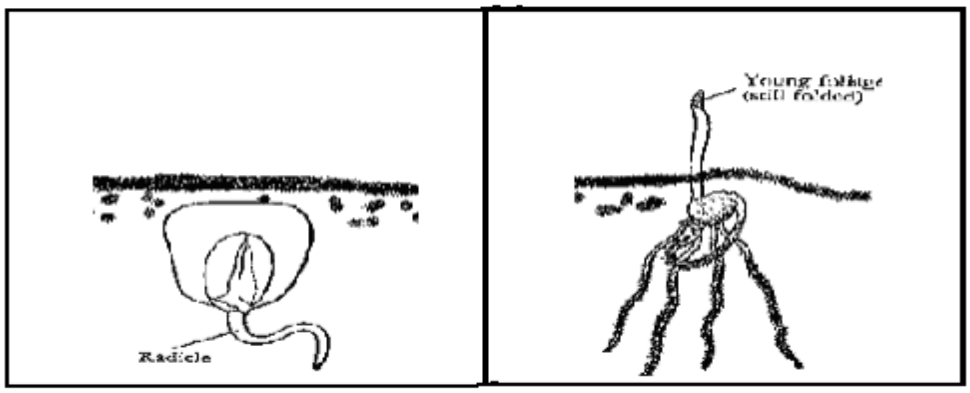
An illustration of hypogeal germination.



Examples of seeds that carry out hypogeal germination

- Maize seeds
- Sorghumseeds
- Millet seeds
- Wheat seeds
- Rice seeds
- Oats seeds
- Mango
- Grampeas
- Kidney beans

Stages in hypogeal germination



Conditions needed for a seed to germinate

. Water or moisture

. Oxygen

. Warmth.

Importance of the conditions needed for germination

a. Water

. It softens the cotyledons.

. It dissolves food in the cotyledon for the embryo to use.

b. Oxygen

. Helps in respiration

c. Warmth

. Provides optimumtemperature for cell activity.

QN :Which part of air is given off during germination

carbon dioxide

Lesson 5

EXPERIMENTS ABOUT THE CONDITIONS NECESSARY FOR GERMINATION

a. Requirements

. .

Seeds

. Ice

. Cotton wool

. Test tubes

. Boiled water

STEPS

- Get four test tubes and mark on them1 -4
- **Test tube i**
- Place a piece of dry cotton wool at the bottomand put in some seeds
- Put the test tube in a warmplace so that seeds can get enough wormth.air but no water

Test tube ii

. Place wet cotton wool at the bottomand some seeds

. Add cool boiled water .then pour a layer of oil on top

Test tube iii

. Put wet cotton wool in the test tube and put in seeds

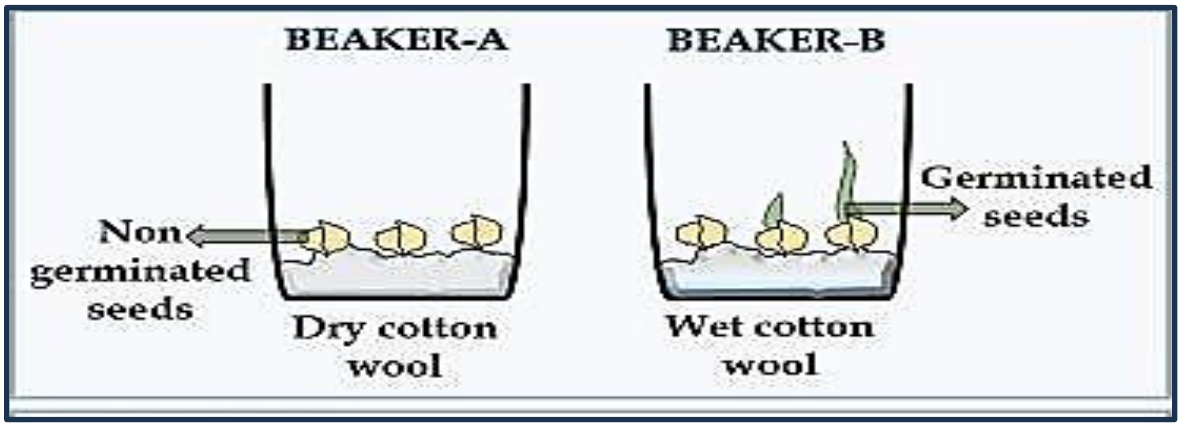
. Place the test tube in a warmplace for seeds to get water,air and warmth

Test tube iv

. Put wet cotton wool with some seeds in test tubes

. Put ice blocks in the test tube

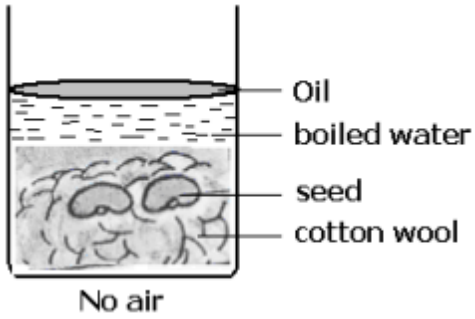
An illustration on conditions needed for germination to take place



a) Testing for water whether it is necessary for germination.

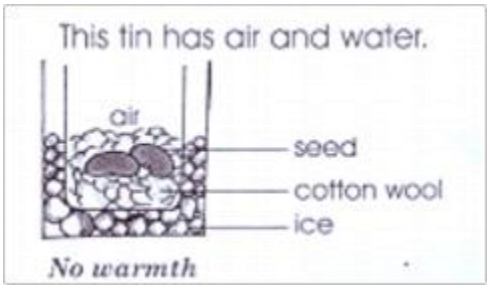
The experiment above has oxygen, warmth without water. So the seeds will not germinate.

b) **Testing for oxygen whether it is necessary for germination.**



The experiment above has water, warmth without air. So the seeds will not germinate.

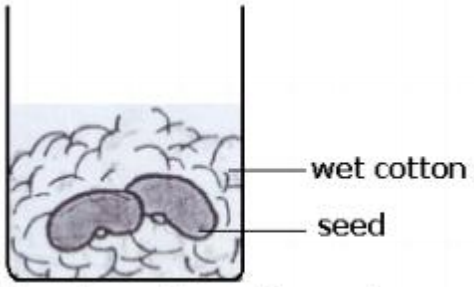
c) **Testing for warmth whether it is necessary for germination.**



The experiment above has water, oxygen without warmth. So the seeds will not

germinate.

d) Testing on germination.



Has air, water and warmth

The experiment above has water, oxygen and warmth. So the seeds will germinate.

OBSERVATION

- After 3 days the seeds in test tubes 3 germinated
- The seeds in the other test tubes did not germinate

CONCLUSIION

- In test tube 3 all the condition were present .
- So, for a seed to germinate, all the conditions must be present.

Lesson 6

Seed viability

This is the ability of seeds to germinate when all the necessary conditions are present.

Experiment for seed viability



Requirements

- clean water
- can
- seeds

Procedure

1. Fill in the can with clean water
2. Drop all the seeds in the can

Observation

Sinking seeds will germinate,floafing seeds will not germinate.

Conclusion

Sinking seeds are able to germinate because they are live

They are not damaged

VIABLE SEEDS

Viable seeds are seeds that can germinate when necessary conditions are present.

Qualities of viable seeds

- Seeds should be mature and dry
- Seeds should be not have holes
- Should not be broken
- Should not be rotten
- Should be of good quality

What is seed dormancy?

Seed dormancy is a condition when the seed does not have the necessary conditions to germinate,

FACTORS THAT LEAD TO SEED DORMANCY

- Broken seeds
- Damaged seeds
- Unhealthy seeds
- Premature seeds
- Holes in seeds

END OF TOPIC REVISION EXERCISE.

1. Give one example of a dicotyledonous seed.

2. What name is given to the following?

   a. Male part of a flower.

   b. Female part of a flower.

3. How is chlorophyll important during photosynthesis?

4. Why does a seed planted in soil which is covered in ice fail to germinate? 5. How are leaves of a bean plant different fromthose of a maize plant?

6. State any one way plants depend on animals.

7. Mention one plant that develops a fibrous root system.

8. What are flowering plants?

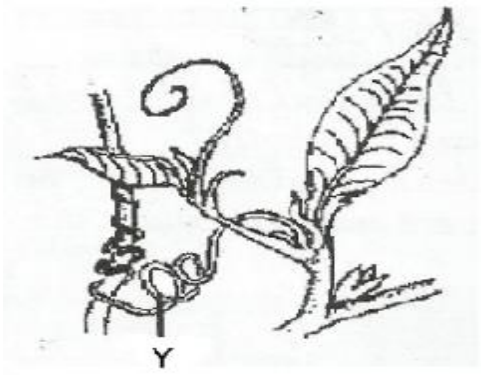
9. State one way plants benefit fromother plants.

10. What is photosynthesis?

11. Mention two things plants need to carry out photosynthesis.

12. What name is given to the main product of photosynthesis?

The diagrambelow shows a part of a plant. Use it to answer questions that follow.



a. What type of stems is shown in the diagramabove?

b. How is part marked Y useful to the stem?

c. Why do plants with weak stems climb others?

d. Apart fromclimbing stems, name any other type of stems possessed by plants. 2. Mention the importance of water to a germinating seed.

3. How is burning similar to germination?

4. Name the pollinating agent that pollinates flowers at night.

WEEK 5

LESSON 1

GROWING CROPS AND FOOD PRODUCTION

GROWING CROPS

Crops are plants grown by man for a purpose

WHY FARMERS GROW CROPS.

. For food

. For income

. For feeding animals

. T0 get raw materials for industries

Types of crops commonly grown

- Fruit crops
- Cereals/grain crops,
- Vegetable crops,
- Legume crops
- Root crops.

Fruit crops

What are fruit crops?

Fruit crops are crops that bear fruits.

Examples of fruit crops

- Apple crops
- Orange crops

- Mangoes crops
- Pineapples crops
- Paw paw crops

Cereals

Cereals are plants that bear grains.

Examples:

- Rice crops
- Wheat crops
- Maize crops
- Oat crops
- Sorghumcrops
- Millet crops

Vegetable crops

They are crops that are eaten as sauce

EXAMPLES OF VEGETABLE CROPS

- Cabbage crops
- Cucumber crops
- Spinach crops
- Lettuce crops

Root crops

These are crops with swollen roots that are eaten.

Examples of root crops

- Cassava crops
- Sweet potato crops
- Carrot crops

- Beet roots
- Root turnip

LESSON 2

Legume crops

These are crops that have root nodules.

Examples:

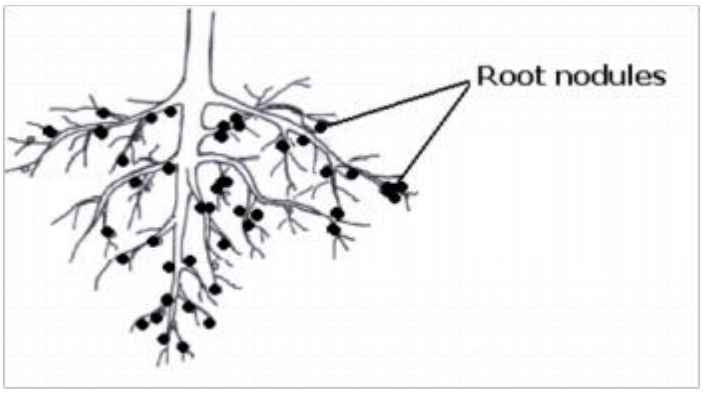
- Bean crops
- Pea crops
- Groundnuts crops
- Soya bean crops
- Simsim
- Cow peas

Characteristics of legumes

. They have root nodules

. They have pods

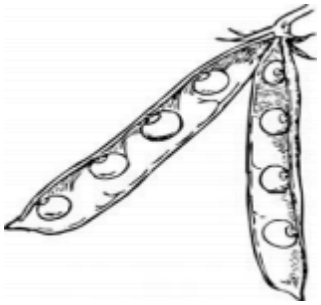
Diagramshowing root nodules.



Note:

1. Root nodules keep nitrogen fixing bacteria.
2. Nitrogen fixing bacteria fix nitrogen in the soil.
3. Nitrogen improves soil fertility.

The structure of a pod



LESSON 3

GROUPS OF CROPS

Crops are grouped according to the time they take to mature,

There are mainly two groups of crops :

- Annual crops
- Perennial crops

Annual crops

- These are crops which mature within a year.
- We harvest themonly once.

Characteristics of Annual crops

- They mature within a year
- Produce yields once in their lifetime

Examples of Annual crops:

- Maize
- Rice
- Beans
- Groundnuts
- Cabbage
- tomatoes
- sorghumetc.

Perennial crops

- These are crops that take more than a year to mature.
- They give yields yearly

Examples Perennial crops :

- Coffee,
- Tea,
- Banana,
- Oranges,
- Sugar cane,
- Mangoes
- Apple
- Jackfruit
- Avocado
- Straw berries etc.

Why do farmers prefer growing annual crops to perennial crops?

Annual crops mature faster than perennial crops.

LESSON 4

GARDEN TOOLS, EQUIPMENT AND MATERIALS

These are simple machines farmers use to carry out some activities in the garden

Examples of garden tools and their functions

1 . **Hoe** -used for digging, weeding and harvesting root crops, planting crops

2. **Panga** - used for cutting small trees, harvesting some crops, pruning trees, clearing weeds

3. **Rake** -used for colle;cting dugout weeds, spreading manure and levelling soil 4. **Watering can** - used for watering crops.

4. **Slasher** - used for slashing weeds

5. **Trowel** - used for digging out seedlings during transplanting.

6. **Wheelbarrow** - used for transporting manure, tools, harvested crops, transporting seedlings.

7. **Axe**-used for cutting big trees

8. **Secateur** - used for pruning.

9. **Garden fork**- used for turning manure. Or lifting manure

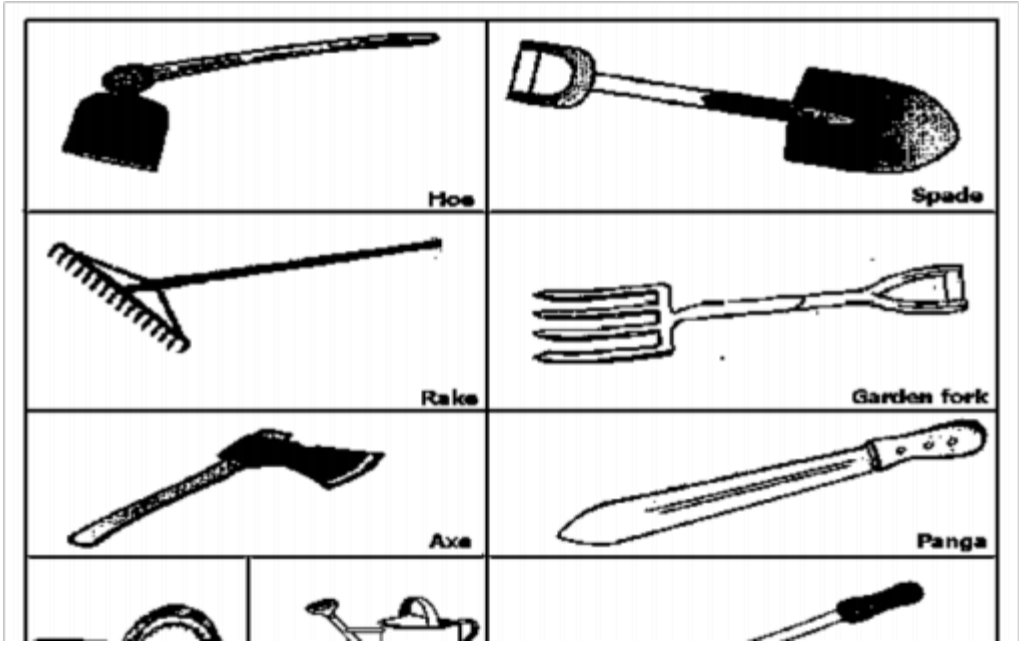
10. **Forked hoe** - used for harvesting rhizomes, dig out deep rooted weeds, digging hard grounds for hollowing

11 . **Sickle** - used for harvesting wheat and rice, cutting grass

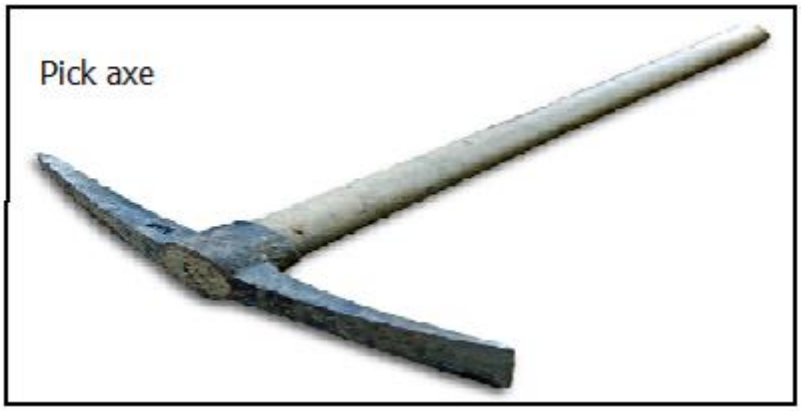
12. **Pickaxe** - used for digging hard grounds and stony area

13-**Knapsacksprayer** –for spraying pesticides and herbicides

Diagrams of garden tools



Caring for garden tool



Caring for garden tools.

- Garden tools must be kept in cool a dry place.
- Cleaning themafter use
- Sharpen cutting tools regularly.
- Replacing spoilt parts

- oiling/greasing moving parts to prevent rusting
- Painting those that need to be painted
- Using garden tools for the right purpose

Ways of preventing garden tools fromrusting

. Painting garden tools

. Oiling garden tools

. Keep garden tools in cool dry places

Effects of rusting on garden tools

. garden tools become blunt

. Garden tools become weak

. Tools develop holes

LESSON 5

FOOD PATH

Food path are different stages of food production

The types of food path

. Village food path

. Town food path

Earning food path

a) Village food path

The type of food path where the food is grown for home consumption.

Stages in village food path

. Land preparation

. Selecting planting materials/ seeds

. Planting

. Caring for crops

. Harvesting

. Storing

. Preparation of food

. Eating

b) Town food path

The type of food path where people in town get food

Stages involved in a town food path

. Harvesting

. Storing

. Transportation of food to the market.

. Selling the food.

. Preparation of food

. Eating

c) Earning food path

The type of food path where food is obtained after earning salary

. Stages in earning food path

. Earning salary

. Budgeting

. Buying food

. Preparing food

. Eating.

Blocks in food path

. These are factors that affect the production of food.

Examples of blocks in a food path

. Pests

. Diseases

. Drought

. Civil wars

. Infertile soil
. Bad roads
. Floods
. Storms
. Hailstones
. Poverty
. Landslides

LESSON 6

CROP GROWING PRACTICES

- Land clearing
- Digging/ ploughing
- Selecting planting materials
- Planting/ sowing
- Caring for crops e.g. weeding, mulching, pruning, thinning etc

Land preparation

it is the way of making land ready for planting

- it'sdone to remove the bush, trees and grass on the soil.
- it'sbetter to clear land in the dry season.

Steps in land preparation

. Clearing land which involves cutting grass and trees removing tree stumps

- Clearing the landis done during the dry season

Digging/ Ploughing

. This is done to make the soil loose, soft and also to burry weeds.

. To allow water and air to enter the soil.

. It is done using hoes, tractors, ox ploughs and forked hoes.

Harrowing the land

- This is the breaking of big lumps of soil which are left after Ploughing.
- It makes the soil loose.
- Also it kills weeds which could have started germinating after Ploughing.
- It is done using a tractor with disc harrows, ox plough, forked hoe and hand hoe.

Selection of planting materials

Planting materials are the different parts of a plant used for plant multiplication.

Examples of planting materials

- Seeds
- Stemcuttings
  - Drought
  - Civil wars
  - Infertile soil
  - Bad roads
  - Floods
  - Storms
  - Hailstones
  - Poverty
  - Landslides
- Vines
- Suckers
- Bulbs
- Stemtubers

Factors to consider when selecting planting materials

. Materials should be free fromdiseases and pests.

. Should not be damaged. broken

. Seeds should be mature and well dried.

. Stemcuttings should bear lateral buds.

. Seeds should not be rotten

WEEK 6

LESSON 1

PLANTING SEEDS

1. Some seeds are planted directly into a well-prepared garden.

Examples

. Maize

. Beans

. Peas

. Cotton

. Groundnuts etc.

1. Others are first planted in a nursery bed.

Examples of seeds first planted in a nursery bed

. Tomatoes

. Onions etc

. Cabbage,

. Eggplants

. Lettuce, spinach etc

Anursery bed

A nursery bed is a small garden prepared for raising seedling

Diagram of a nursery bed.



Note: Seeds of these crops cannot grow through heavy soils.

Site for a nursery bed

- Should be in a place protected fromdirect wind.
- A place with good drainage.
- A place near a water source.
- A place far frombushes/forest
- A place protected fromanimals.

Activities carried out on a nursery bed

. Watering
. Spraying with chemicals
. Weeding
. Hardening off
. Thinning

Note:

Hardening off is a way of making seedlings get used to conditions they are to meet in the garden.

Importance of a nursery bed

a) It protects the seedlings fromharsh weather conditions e.g.

i. Direct sunlight

ii. Direct raindrops

iii. Strong wind

b) It helps in proper selection of good seedlings.

c) It gives farmers time to prepare the main garden.

d) It helps in providing easy care to the seedlings

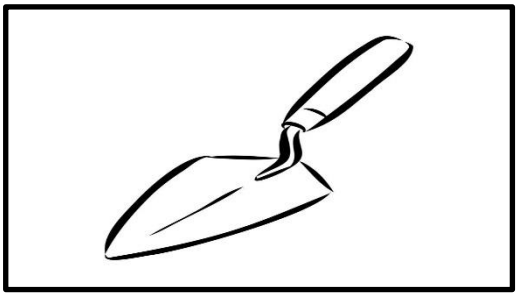
Transplanting

. Is the transfer of seedlings fromthe nursery bed to a well prepared main garden

. It is done in the evening to avoid excess loss of water through transpiration

. To avoid wilting

Trowel



Why is atrowel used for transplanting?

. It scoops seedlings without damaging the roots

Time/Season of planting

. Planting should be done during the wet season.

. It is good to plant early during the first rains because:

Importance of early planting

o Crops make proper use of rainfall for that season.

- Crops grow fast enough and compete well with weeds.
- Helps to control pests and diseases.
- Crops mature fast and get good market.

LESSON 2

Methods of planting seeds.

- Row planting
- Broadcasting

Row planting

This is the planting of seeds/ seedlings in a line.

Examples of seeds planted using row planting:

- Beans
- Maize
- Groundnuts
- Cotton
- Soya beans
- Coffee
- Tea

Advantages of row planting

. Few seeds are used.

. It is easy to weed the crops.

. Spacing in lines helps to control pests.

. It is easy to spray the crops.

. It is easy to harvest the crops.

Disadvantages of row planting

. Takes a lot of time/tiresome.

. It is tiresome/tiring

Broadcasting

This is when seeds are thrown randomly.

Examples of seeds planted using broadcasting:

o Millet

o Sorghum

. Sim-sim

. Rice

. Beans

. Maize

Advantages of broadcasting

. Time saving.

. Requires less labour force

Disadvantages of broadcasting

. It wastes seeds.

. It is difficult to weed,

. it is difficult to spray

. It is difficult to harvest

. It can lead to overcrowding of seedling

LESSON 3

FARM PRACTICES

Gap Filling

1. This is the replacement seeds or seedlings in spaces where they didn'tgerminate. 2. It is done not to waste land.

Thinning

Thinning is the removal of excess or poorly growing seedlings fromthe garden

Examples of crops that need thinning

- Maize,
- Cotton,
- Rice,
- Millet,
- Sorghum, etc.

Methods of thinning:

- Uprooting/
- Cutting the plant at the base area.
- Digging out the plant

Importance of thinning

- Thinning controls pests .HOW by reducing hiding places for pests.
- It reduces competition for sunlight, minerals and water among crops.
- Thinned materials can be used for mulching.
- It improves crop yields
- Thinned materials can be used for gap filling.

Pruning

Pruning is the removal of excess branches or leaves froma plant

Garden tools used for pruning

- Secateurs
- Pruning saw
- Shears

Importance of pruning

- It controls crop pests.
- Reduce weight on the plant.
- Reduce competition for sunlight,
- It improves on crop yields
- It makes harvesting easy
- Pruning gives good space for harvesting.

Mulching

1. Mulching is the covering of soil with dry plant materials.

Plant materials that can be used:

- Dry maize stalks
- Dry leaves
- Coffee husks
- Woodshaving
- banana fibres
- Dry grass

Advantages of mulching

- It maintains soil moisture
- Mulching controls soil erosion.
- Mulching improves soil fertility
- Mulching controls the growth of weeds.

Disadvantage of mulching

. Mulches hide pests.

. Mulches can easily catch fire and destroy crops.

. It is tiring to mulch a large garden

LESON 4

Manuring

. Act of adding manure to soil.

. Manures is natural fertilizers.

. Manure is got fromdead plants, animals and animals dung.

. Manure helps to improve soil fertility. **Types of manure**:

. Compost manure made fromkitchen refuse and plant remains

. Farmyard manure made fromanimals dung and urine and animal waste

. Green manure made fromgreen plants when ploughed into soil

. Organic mulches made fromrotting of mulches

Watering

. Act of providing water to crops when there is little water in the soil.

. It is done to prevent crops fromdrying and dying.

. Watering can also be done to newly transplanted seedlings.

. It is done using a watering can.

. It is done late in the evening.or early in the morning

weeding

- Weeding is the removal of unwanted plants fromthe garden.
- Weeds are plants growing where they are not wanted

Examples of common weeds:

. Wandering jew
. Spear grass
. Blackjack
. Star grass
. Wild finger millet
. Couch grass
. Elephant grass
. Pig weed

Importance of weeds

. Weeds are source of food for both people
. Legume weeds improve on the soil fertility by adding nitrogen. Controlling soil erosion

Weeds are used for feeding animals

Dry weeds can be used as mulches

Dangers of weeds

. Weeds hide pests .
. Weeds compete with crops for sunlight, soil nutrients and water with crops.
. Weeds make harvesting difficult
. Weeds contaminate yields weeds lower the quantity of harvests
. Weeds make pruning difficult
. They costly to manage
. Some are poisonous to animals.

Importance of weeding

. Weeding controls crop pests
. It makes harvesting easy

. It makes spraying easy

. It makes pruning easy

. It reduces completion for sunlight, soil nutrients and water between crops and weeds

Ways of controlling weeds

. Spraying with herbicides.

. Mulching.

. Uprooting.

. Slashing (slashing is normally carried out in crops with wide spacing e.g. Coffee and banana plantation)

. Digging themout

LESSON 5

CROP PESTS

They are organisms that destroy crops

COMMON CROP PESTS

	Crops	Pests		Affected part
1	Cassava	Monkeys baboons Mole rats		Roots
2.	Sweet potato	a	Sweet potato weevil	Tuber
		b	Squirrel	Roots
		C D	Eel worms Mole rats Monkeys baboons	Roots
3.	Irish potato	Irish aphid		Leaves
4.	Carrots	Eel worms		Roots
5.	Sugar cane	a. Stalk borers		Stems
		b. Aphids		Stems
6.	Banana			Fruits
				Stem(corm)
9	Cabbage			Stem

10	Coffee	locusts	Leaves
11.	Maize	Maizestalk borer Weaver birds Termites Rats weevils	Stems Grains

LESSON 6

COMMON DISEASES OF CROPS

	Crop	Diseases	Affected part
1	Cassava	a Cassava mosaic	Leaves
		b Brown streak	Leaves and tubers
2	Sweet potato	Sweet potato blight	Leaves and stem
3	Coffee	a Coffee berry disease	Coffee beans
		b Wilt disease	Leaves
	Banana	a Panama disease	Leaves and stem
		b Wilt disease	Leaves
			leaves
6	Sugar cane	Ratoon stunting disease	Stem
7	Groundnuts	Rosette disease	Leaves

LESSON 6

COMMON DISEASES OF CROPS

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1	Cassava	a Cassava mosaic	Leaves
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	Banana	a Panama disease	Leaves and stem
		b Wilt disease	Leaves
			leaves
6	Sugar cane	Ratoon stunting disease	Stem
7	Groundnuts	Rosette disease	Leaves

8	Maize	Maize streak rust	Leaves

signs of pest damage to plants

. Holes in leaves, seeds, and roots

. Spots on leaves

- Deformed part .
- Rotten plant parts

Signs of disease damage to plants

- Deformed parts
- Premature ripening of fruits
- Stunted growth

Methods of controlling pests

- Uprooting
- Proper spacing
- Early planting

Methods of controlling crop diseases

- Uprooting infected crops
- Proper spacing
- Early planting
- Spraying with pesticides
- Using scarecrows
- Setting trap
- Crop rotation
- Planting goodse
- Rearing cats

UPROOTING INFECTED CROPS	USE OF PESTICIDES/SPRAYING
	
CROP ROTATION	**USING SCARE CROWS**
	
USE OF TRAPS	
	

WEEK 7

LESSON 1

Harvesting of crops

- Harvesting is the collecting or removal of ready crops froma garden.
- Harvesting should be done when the crops are ready.
- Most crops that need drying should be harvested in dry season.

NOTE:

Harvesting is done during the dry season because there is enough sunshine to dry the harvested crops.

Methods of harvesting crops

A. Cutting

Examples of crops harvested by cutting:

- Banana
- Sugar cane
- Millet
- Sorghum
- Wheat
- Rice
- Maize

B. Picking/Plucking

Examples of crops harvested by picking

- Coffee
- Tomato
- Eggplants
- Cocoa

C. Uprooting using hands

Examples of crops harvested by uprooting using hands

. Groundnuts

. Onions

. Carrots

. Cassava

D. Digging out the crop

Examples of crops harvested by digging out the crop

. Cassava

. Sweat potato

. Yams

. Coco yam

. Irish potato, etc

LESSON 2

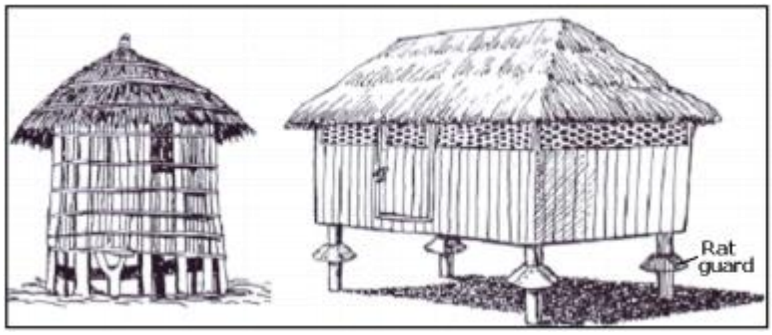
Storing harvested crops

1. Harvested crops are commonly stored in granaries, stores, silos, maize cribs, etc.

. Crops that can rot easily can be stored in special made structures with coolers.

. Crops that do not rot easily can be stored in granaries, silos, maize cribs and stores.

Granaries



Conditions for proper storage

. Harvested crops should be stored when they are dry.

. The roofs of stores should not leak.

. Stores should have good ventilation.

. Rat guards should be fixed on the granary.

. The seeds should be dusted with pesticides to protect themfrompests.

Importance of storing harvested crops.

. It prevents wastage.

. Helps to get what to eat in the dry season.

. Helps farmers to sell their produce when there is good

. It helps farmers to keep what to plant in the next season.

LESSON 3

Food preparation

What is food preparation?

Food preparation is the making of food ready for eating.

Give any four ways of preparing food

- By roasting
- By baking
- By boiling
- By frying
- By deep frying
- Steaming

Why do people prepare food?

- To make food soft
- To kill germs and worms
- To make food tasty

LESSON 4

FARMRECORDS

Farmrecords are written information about activities carried out on the farm

Examples of Records kept by crop farmers:

- Sales records,
- Production records
- Health records
- Expenditure records

Uses of farmrecords

. Records help the farmer to know his profits or losses.

. helps to make decisions

. Records help the farmer to plan for the farm

. for proper taxation of the farm

. Records help the farmers to get loans.

. It helps to know his income and expenditure

END OF TOPIC REVISION EXERCISE.

1 . To what group of crops does a sweet potato belong?

2. State one way root nodules are important to soil.

3. State one danger of leaving garden tools exposed to water and oxygen.

4. Mention one importance of mulching crop gardens.

5. What is a pest?

6. Give one example of apest ina garden.

7. How are pests dangerous to a crop garden?

8. Give the importance of each of the following garden tools to a farmer:

a. Wheelbarrow

b. Hoe

c. Garden trowel

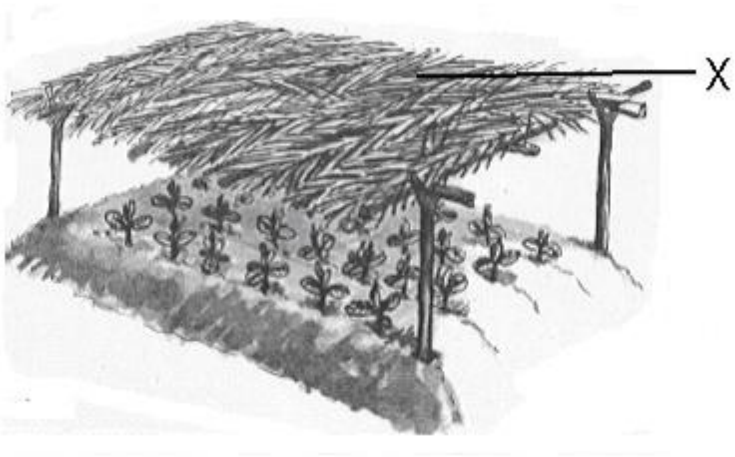
9.How are the following practices important in crop growing?

d. Pruning

e. Weeding

f. Thinning

11. **Use the diagrambelow and answer questions that follow.**



a. Name part marked X on the diagram.

b. Name the structure shown in the diagram.

c. How is the structure useful to a farmer?

d. How useful is part marked X in the structure above?

e. Identify any one activity carried out in the structure above.

12. Give one example of manure.

13. Mention one way aP.4 child can identify a leguminous crop.

14. Why is harvesting of cereals done in dry seasons?

15. Name one crop propagated by stemcuttings.

16. Give one method of planting seeds.

LESSON 5

CHANGES IN WEATHER AND CLIMATE IN THE ENVIRONMENT

Weather

Weather is the condition of the atmosphere of a place at a given/ particular time.

Types of weather

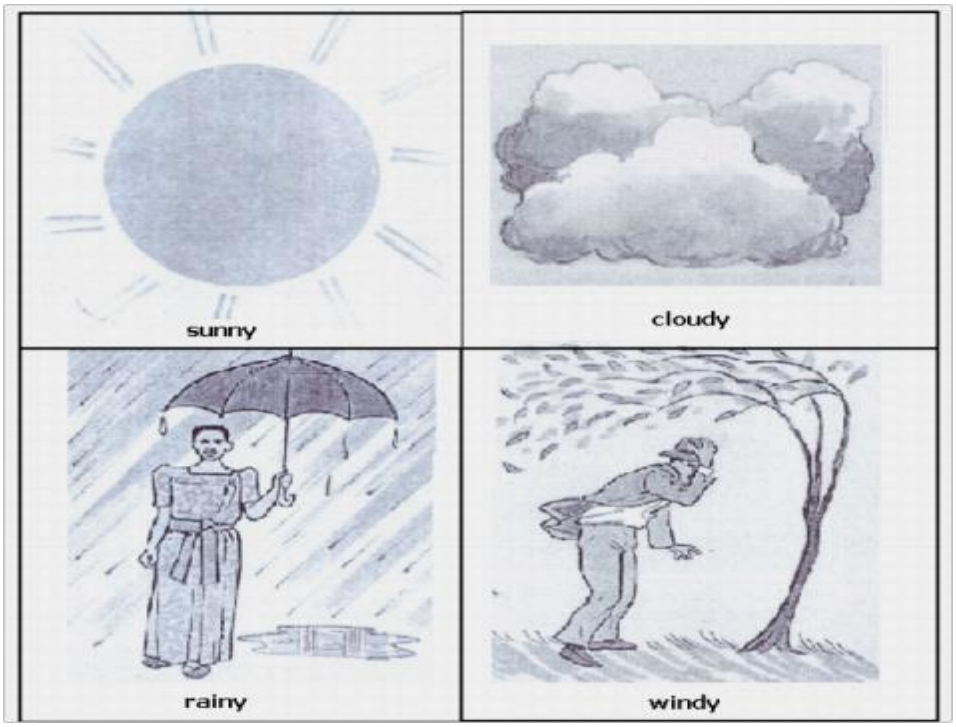
. Rainy weather

. Windy weather

. Sunny weather

. Cloudy weather etc

AWEATHER CHART



Elements of weather

- Rain
- Cloud cover
- Sunshine
- Temperature
- Humidity
- Air pressure/ Atmospheric pressure

LESSON 6

CLOUDS

. Clouds are condensed masses of water vapour floating in air.

. Clouds are formed when the rising water vapour condenses in the atmosphere.

. Clouds are grouped according to their heights and general shape.

Types of clouds

. Cirrus clouds.

. Stratus clouds.

. Nimbus clouds.

. Cumulus-nimbus clouds

Nimbus clouds

. These are clouds that bring or give us rain.

. They don'thave any special shape.

. They are the nearest clouds to the earth.

Stratus clouds

- They spread in the sky widely in calmflat layers.
- They show a sign of bad weather.

Cumulus clouds

- They are white clouds, which resemble cotton piles
- They can develop into thunderclouds and thus they indicate rain.

Cirrus clouds

- They look like feathers in the sky.
- They are the furthest in the sky.
- Cirrus clouds appear in dry weather and often indicate storm.

Cumulonimbus

- They are observed during storms
- They cause thunderstorms
- They are close to the ground
- They contain a lot of moisture

Advantages of clouds

- Clouds protect us fromdirect sunshine.
- Clouds give us rain

Dangers of clouds

- Cause lightning
- Bring heavy rainfall
- Cause accidents to people who use air transport

How are clouds formed?

By condensation process

WEEK 8

LESSON 1

WATER CYCLE.

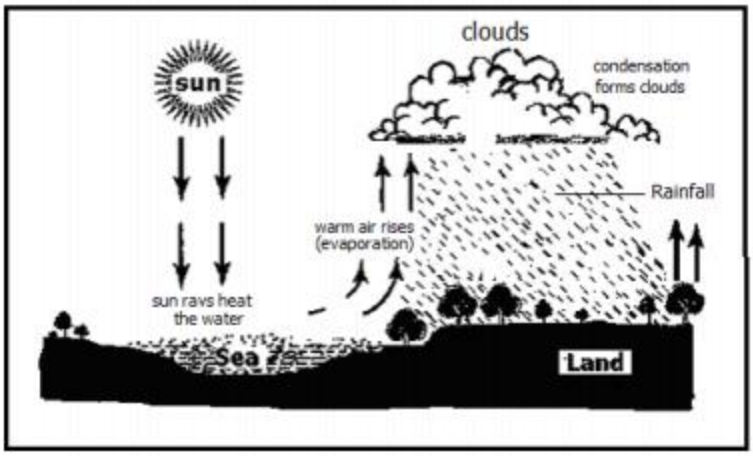
. A water cycle is a process by which rain is formed.

. Heat fromthe sun causes evaporation of water in the water body and transpiration in plants

. The vapour condenses to formclouds

. The clouds give us rain

A diagramto illustrate a water cycle



Components of the environment that help in rainformation

. Sun

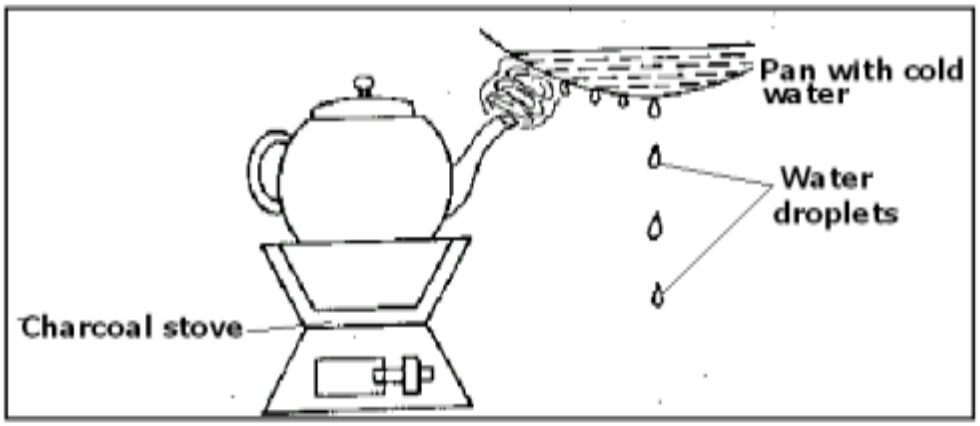
. Plants

. Water bodies

An experiment to show the water cycle

- Make a charcoal stove and make sure that it produces or gives enough heat.
- Put water in a kettle and cover the top outlet leaving only the spout open.
- Put the kettle full of water on a charcoal stove and leave it to boil.
- When water boils, it will give out steam. (Water vapour) through the spout.

e. Get a pan full of cold water and put it near the spout where water vapour comes From



Explantation

. A charcoal stove represents the sun.

. Kettle of boiling water represents the water bodies.

. Steamfromboiled water in the kettle represents water vapour fromwater bodies after

. The sun has heated the water.

. A pan full of cold water represents condensation point.

. Condensed vapour on the surface of a cold pan represent clouds.

. Water droplets represent rain.

LESSON 2

Rain

Is water drops falling fromclouds

Raindrops

Its the total amount of rain received in an area

Importance of rain

- Rain gives us water for cooking, bathing etc.
- Rain-removes dust in space and we breathe in cool air.
- Rain softens the soil.
- Rain adds water to water bodies.
- Rain supports plant growth

Disadvantage of rain

- Too much rain brings floods which destroy property and lives.
- During rain, lightning can kill people and animals.
- Too much rain leads soil erosion.

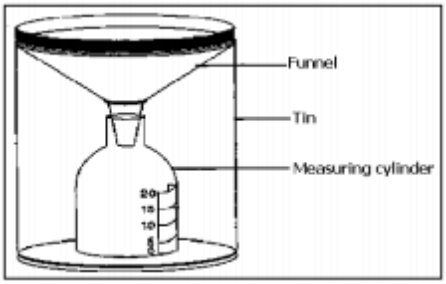
ARAIN GAUGE

- A rain gauge is an instrument used to measure rainfall received in an area.
- It is put in an open space to collect the right amount of rain received.
- Rainfall is measured in millimetres.

Why is rainfall measured in millimetres?

To measure the depth rain has entered the soil.

A diagramof a rain gauge



Sources of water

Natural sources of water

- Rain (main source)
- Lakes
- Oceans
- Seas
- Streams
- Rivers
- Springs

Artificial sources

- Boreholes
- Wells
- Valley dams
- Ponds

Uses of water in the environment

- People use water for bathing, washing plates and clothes etc.
- Water animals get their food and oxygen fromwater.
- Flowing water helps to produce hydro- electricity.

. Water is used for watering crops.

. Animals drink water, etc

Ways water sources get contaminated

. Disposing faeces / rubbish

. Swimming in water sources

. Spitting in water sources

. Washing in water sources

. Urinating in water sources

. Fetching water using dirty containers

Containers for harvesting water

. Saucepan

. Drums

. Basins

. Buckets

. Jericans

. Pots

Maintenance for water sources

. Fencing

. Repairing damaged parts

. Putting strict laws

. Planting grass around

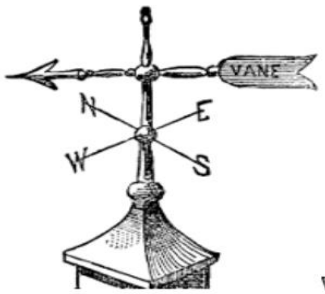
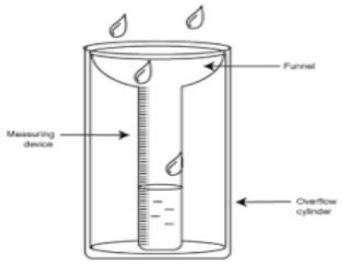
LESSON 3

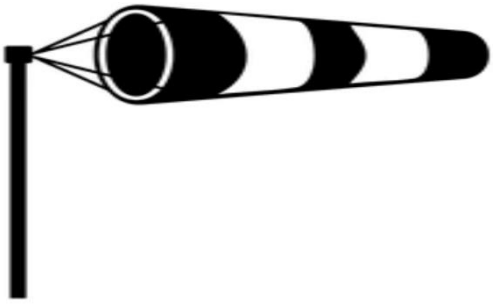
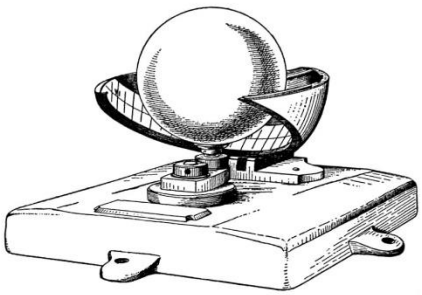
WEATHER INSTRUMENTS

- Weather instruments are found at a weather station.
- At a weather station, delicate instruments are kept in a white box called a Stevenson screen.
- These instruments include the barometer, hygrometer and maximumand minimum Thermometer.
- The sunshine recorder, wind vane, windsock, rain gauge and anemometer are placed in the open space of the weather station.

Weather instruments and their uses

1. Thermometer - Measures temperature of a place
2. Sunshine Recorder - Records hours it has shined in a day.
3. Wind vane - Shows the direction of wind.
4. Windsock - measure the strength of wind.
5. Anemometer - Measures the speed of wind
6. Hygrometer - Measures humidity.
7. Barometer - Measures atmospheric pressure.
8. Rain gauge - Measures rainfall received in an area.

Weather instruments	Pictorial representation
Anemometer... Close Up!  Anemometer	 Wind vane
 Rain gauge	 Barometer
 Thermometer	 Hygrometer

 **Wind sock**	 Sunshine recorder

LESSON 4

TEMPERATURE

- Temperature is the degree of hotness or coldness of a body or place.
- The instrument used to measure temperature is called thermometer.
- A thermometer is read in two scales namely:
- Fahrenheit.
- Centigrade or Celsius.
- Temperature is measured in degrees centigrade or degrees Fahrenheit.

What are the basic units for measuring temperature?

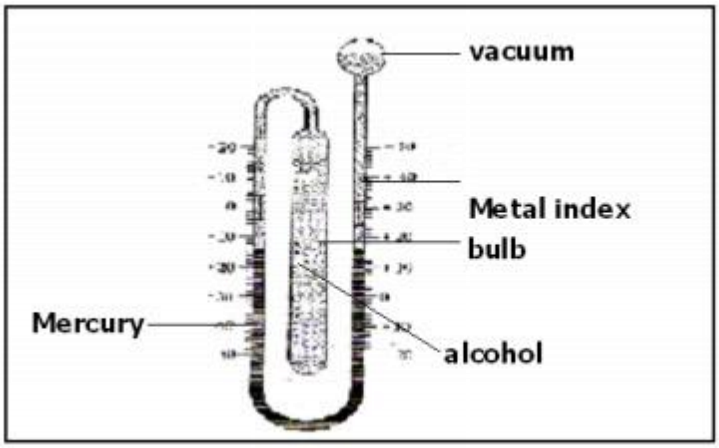
Types of thermometers

1. Clinical thermometer.
2. Minimumand maximumthermometer.
3. Scientific thermometer
4. Wall thermometer

Minimumand maximum/Six'sthermometer

- This is sometimes called the six'sthermometer because it was first made by James six.
- The minimumand maximumthermometer is used to measure the lowest and highest temperature of the day.
- It uses both alcohol and mercury.

The minimumand maximumthermometer



To read the temperature looks at the lower ends of each index.

Clinical thermometer:

1 .How is a clinical thermometer important to a doctor?

It is used for measuring human body temperature

2.What is the normal body temperature for a human being?

37^0 C.

3.State any two body parts where a clinical thermometer is put when taking up the readings.

Under the tongue, under the armpit, in the anus, in the vagina

4.Why is the bulb of the thermometer put under the tongue when taking up the readings?

Under the tongue has constant body temperature.

5.Give any two groups of people who commonly use clinical thermometer

Doctors, Nurses, midwives and first aiders

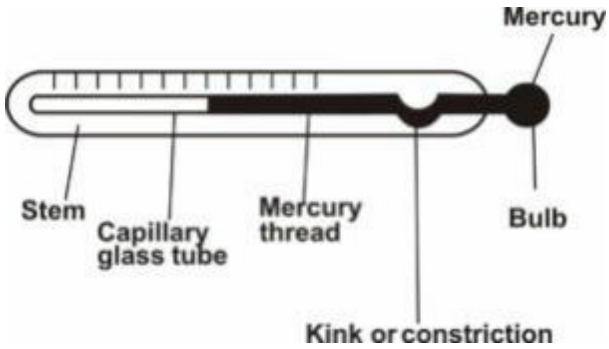
6.Why does the doctor shake the thermometer before using it on another person?

. To reset the thermometer

. To force the mercury togo back to the bulb

. To get correct measurements

7.Draw the clinical thermometer and name its parts.



Functions of parts of a clinical thermometer

Kink: To prevent the back-flow of mercury when taking up readings

Why is the stemmade up of a glass?

The glass is transparent

Why is mercury commonly used in thermometers?

. Mercury is a good conductor of heat.

. Mercury does not stick on the bore.

. Mercury expands uniformly.

LESSON 5

SUNSHINE

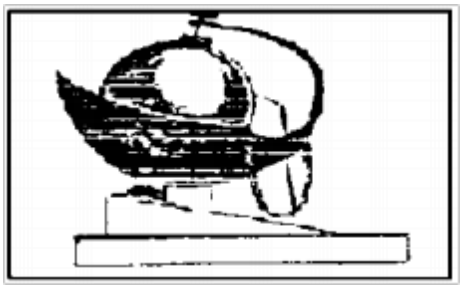
.When the sun is in the sky around mid-day the condition is not the same as it is in the morning or evening.

.Sunny weather is highly determined by the brightness or less brightness or non- existence of sunshine.

.The length of time it has shone on a particular day in a particular place is measured by an instrument called sunshine recorder.

Note: the sunrises in the East and sets in the West

Asunshine recorder



Advantages of sunshine

. Sunlight helps our crops to make their own food.
. Sun heat dries our harvested crops.
. Sun heat helps in the formation of rainfall.
. Sunlight helps our skins to make vitamin D.
. Sunshine helps to kill some germs.

Disadvantages of sunshine/ sun

. Prolonged sunshine makes the day very hot.
. Prolonged sunshine dries crops.
. Prolonged sunshine dries same water sources.

. It causes draught

Effects of the changes in the environment.

. Too much rain causes floods.

. Drought leads to very high temperature in the atmosphere.

. Drought leads to weathering of vegetation leading to lack of food and water for animals.

. Strong wind destroys plants and buildings.

. Too much rainfall leads to soil erosion, etc

LESSON 6

Wind

Wind is moving air or wind is air in motion

Instruments that describes wind and thier diagrams

A. Windvane

It shows direction of wind

Diagramof wind vane



B. Windsock

It measures the strength ot wind

Diagramof windsock



C. Anemometer

It measures the speed of wind

Diagramof Anemometer



. Humidity is the amount of water vapour in the atmosphère at a given time

. Hygrometer is an instrument used to measure humidity.

. Barometer is an instrument used to measure air pressure

END OF TOPIC REVISION EXERCISE.

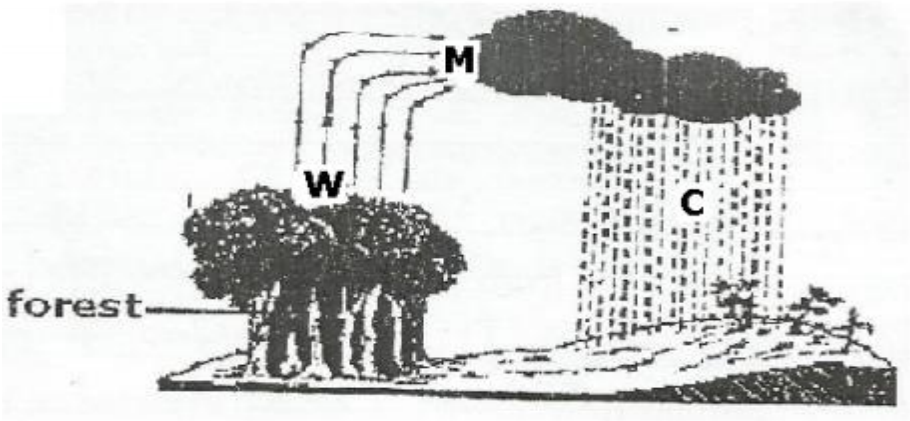
. What is weather?

. Name any one process involved in thewater cycle.

. Name the instrument used to measure humidity in the atmosphere.

. How are nimbus clouds important in the environment

The diagramshows natural changes.use it to answer the questions that follow.



i. What process in nature is shown in the diagram?

ii. Name the process marked W.

iii. Mention the process that takes place at M.

iv. How is C important to plants?

2. Identify any two types of weather.

3. Name the type of clouds that spread in the sky widely in calmflat layers. 8. Why is a rain gauge placed in an open space?

9. Identify any one artificial source ofwater.

10. Give one use of water in a home.

11. Name any one instrument kept in a stevenson screen.

12. What is temperature?

13. The diagrambelow shows one of the instruments used in the community. Use it to answer questions that follow :



a. Name the instrument shown in the diagram.

b. How is the instrument useful to people?

c. State two advantages of sunshine.

WEEK 9

LESSON 1

PERSONAL HYGIENE

. Personal hygiene is the general cleanliness of the body

. It is the keeping of the body clean.

Parts of the body that needs proper cleaning:

. Fingers/Finger nails

. Hair

. Face

. Teeth

. Armpits

. Sexual organs

. Between the buttocks

LESSON 2

Ways of keeping our body clean

- Bathing using safe water and soap.
- Cutting finger nails short.
- Washing the face after sleep.
- Washing the hands whenever they are dirty.
- Shaving hair.
- Combing hair using clean combs.
- Washing clothing and beddings.
- Brushing teeth regularly.
- Brushing shoes

Reasons why we washour hands

- To remove germs
- To washaway germs
- To prevent diarrhoea or faecal diseases

LESSON 3

Things used for keeping our bodies clean

- Bathing sponge
- Safe water
- Towel
- Soap
- Shavers
- Combs
- Tooth brush and tooth paste
- Razor blades
- Nail cutters

. Dental floss

LESSON 4

Importance of keeping our bodies clean

. To prevent bad smell on the body.
. To prevent diseases.
. To be healthy.
. To remove sweat
. To kill germs
. To kill parasites.

Keeping beddings and clothing clean

. Washing themusing water and soap/detergents.
. Ironing themwhen washed.
. Spraying disinfectants on them.
. Keeping themin clean safe places if not in use.

Why we iron

. To kill germs
. To remove creases or folds
. To kill parasites

Things used to clean beddings and clothing.

. Soap/ Detergents
. Washing brushes
. Water

Importance of keeping beddings and clothing clean

. It prevents the breeding of lice and bedbugs

. It prevents bad smell

. It prevents skin diseases

LESSON 5

Why we do body physical exercises

.To befit

.To be healthy

.To reduce fats

.To relax the mind

.To strengthen the muscles

.To reduce weight

END OF TOPIC REVISION EXERCISE.

1 . What is personal hygiene?

2. Give one activity aP.4 child can carry out to promote personal hygiene.

3. Identify one part of the body that needs proper cleaning.

4. Mention any two advantages of keeping the body clean.

5. Besides personal hygiene, name any other element of primary healthcare.

6. Identify any one thing used for keeping our bodies clean.

7. Identify any one major step involved in keeping beddings and clothes

END

PERSONAL HYGIENE

Personal hygiene is the cleanliness of ones self and the things he/she uses.

Parts of the body that needs proper cleaning:

a) Fingers/Finger nails
b) Hair
c) Face
d) Teeth

Things used to clean the body:

a) Clothing
b) Beddings
c) Tooth brush
d) Shoes etc

Importance of keeping our bodies clean

a) Remove dirt.
b) Remove germs.
c) Avoid bad smell.
d) To be healthy.
e) To make the body fresh.
f) Prevent breeding of disease vectors
g) To remove sweat.

Ways of keeping clean

a) Bathing using safe water and soap.
b) Cutting finger nails short.
c) Washing the face/eyes after sleep.
d) Washing the hands whenever they are dirty.
e) Shaving old hair.
f) Combing hair using clean combs.
g) Washing clothing and beddings.
h) Brushing teeth regularly.
i) Ironing washed clothes and beddings.
j) Brushing shoes.

Things used for keeping our bodies clean

a) Bathing sponge
b) Safe water
c) Towel
d) Soap
e) Bathing sponge
f) Shavers
g) Combs
h) Tooth brush and tooth paste
i) Razor blades
j) Nail cutters
k) Dental floss

Keeping beddings and clothing clean

a) Washing them using water and soap/detergents.
b) Ironing them when washed.
c) Spraying disinfectants/perfumes on them.
d) Keeping them in clean safe places if not in use.

Things used to clean beddings and clothing.

a) Soap/ Detergents
b) Washing brushes
c) Water
d) Flat iron/ Iron box
e) Suit cases

Importance of keeping beddings and clothing clean

a) To remove dirt.
b) To remove germs.
c) To remove eggs of disease vectors.
d) To keep away bad smell.
e) To look smart.
f) To prevent disease vectors from breeding.
g) Ironing removes wrinkles and kills eggs of parasites from clothing and beddings.

ACTIVITY;

1. **Collecting all the materials used in personal hygiene.**
2. **Demonstration on how to keep personal hygiene.**

Exercise;

1. **Name any two items a pupil in primary four can use to clean her/himself.**
2. **What is the danger of poor personal hygiene to an individual?**
3. **Give any one disease a P.4 child can get due to poor personal hygiene.**
4. **How do we call the act of keeping oneself clean?**
5. **Give any two things a P.4 child can do to keep him/herself clean.**
6. **Give any two values of maintaining good personal hygiene**.

END